



Data Warehouses

Lecture 10

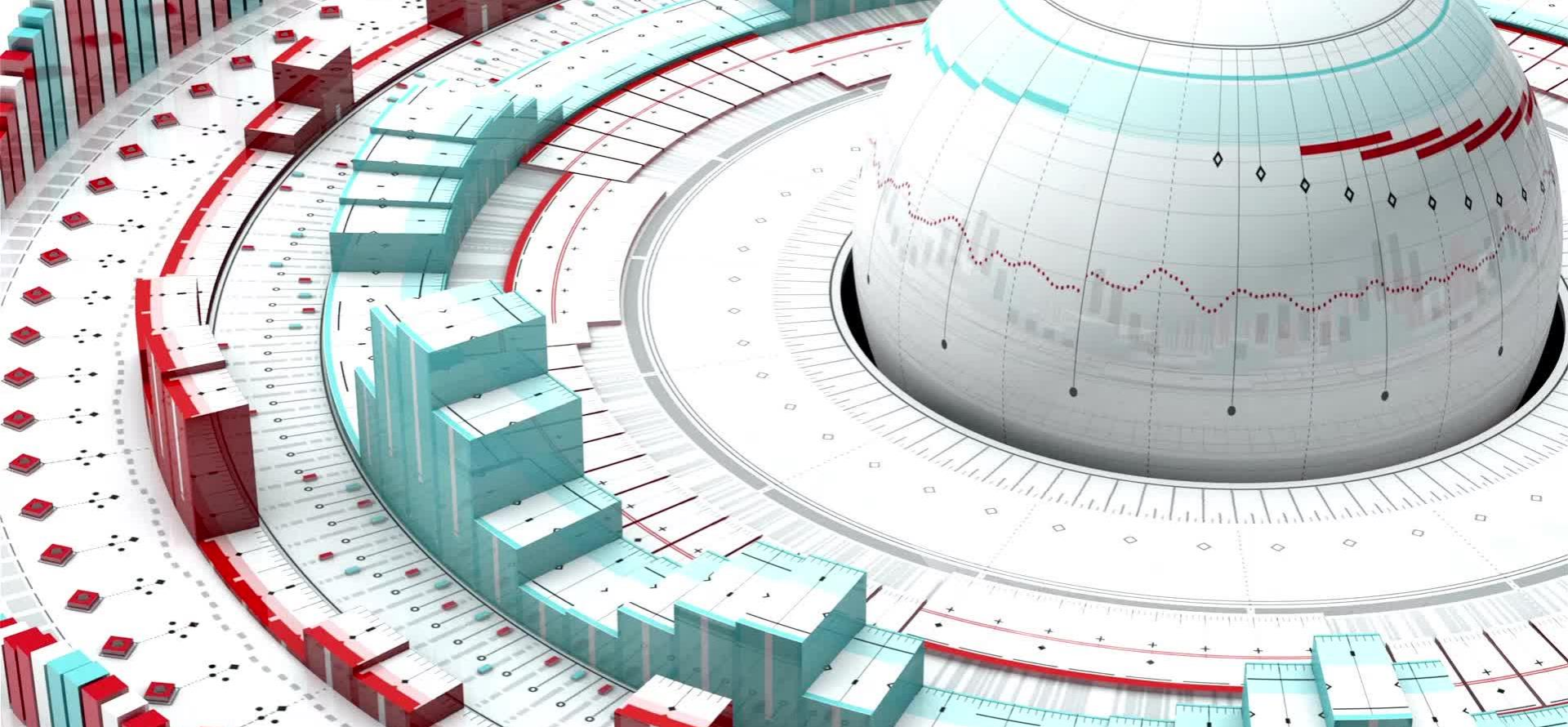


Lecture Goals

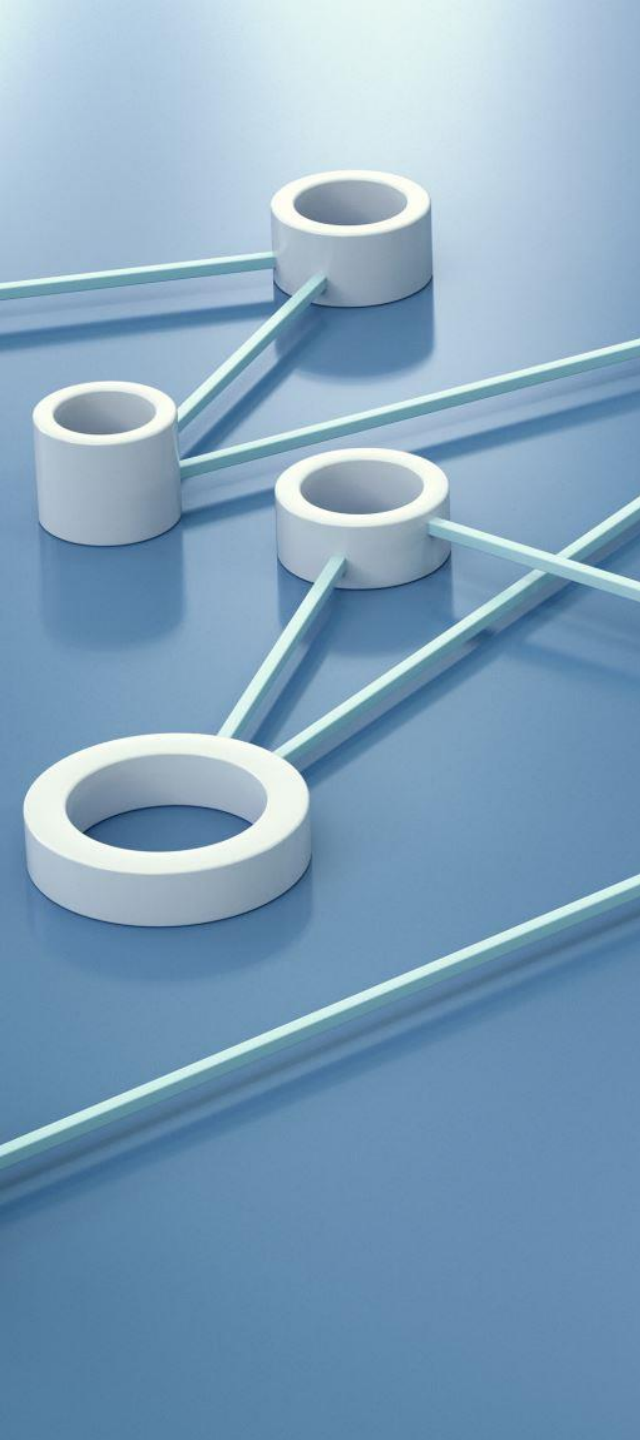
Goal:

Basics of Dimensional Design

Logical data map



DW Design



DW Design

Datawarehouse is a system

Follow systems design approach

As a reminder – decision support system

helps in the decision-making process by using available data from within the company.

In this context

DW is a result of solving questions around specific subjects.

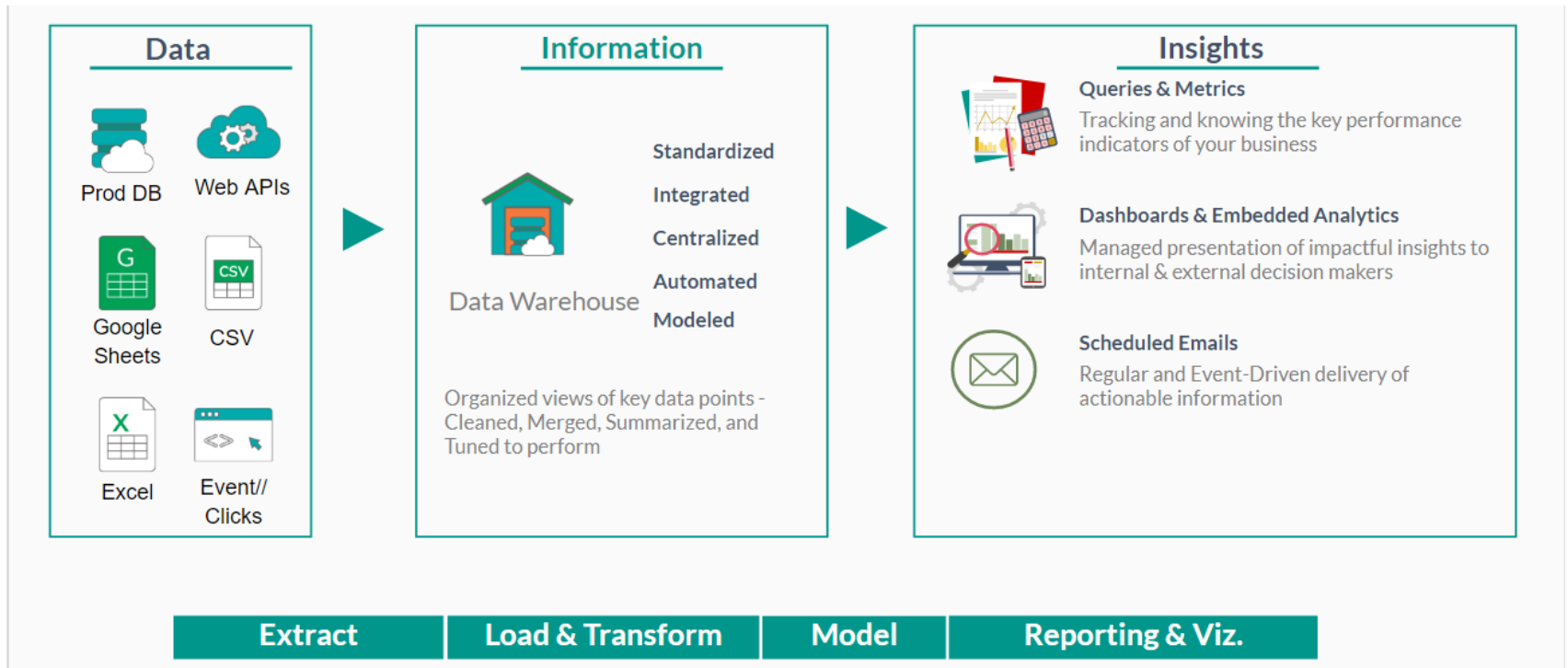
DW project is of a broader scope

Yet, in principle, it is still a design of an information system

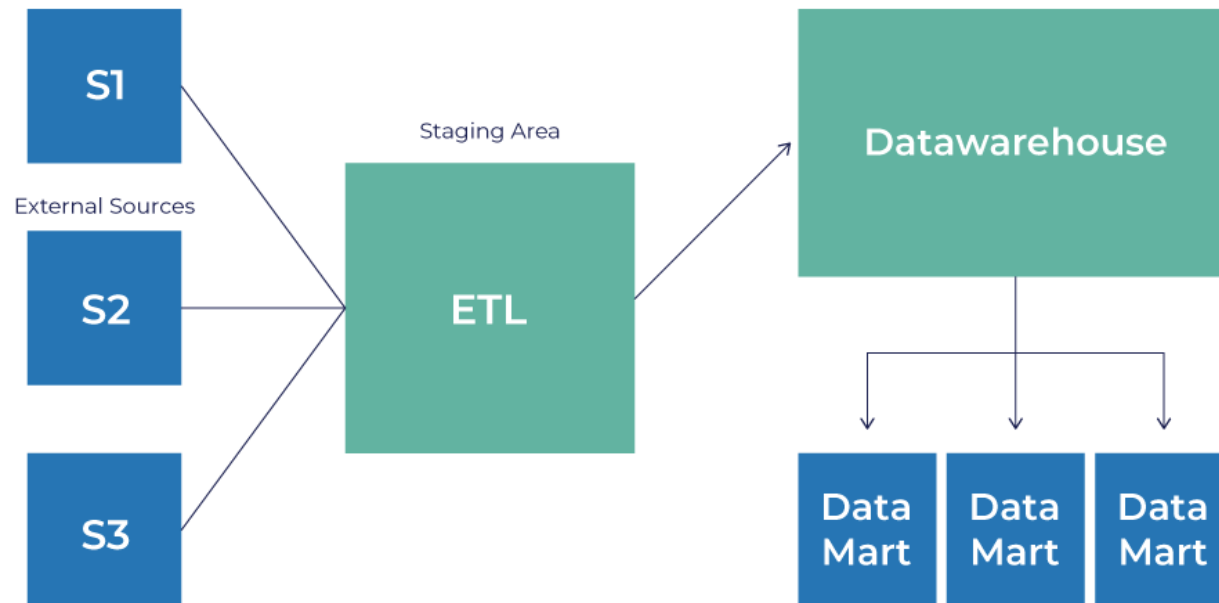
Due to DW systems' peculiar characteristics

different approach from the typical operational applications is needed

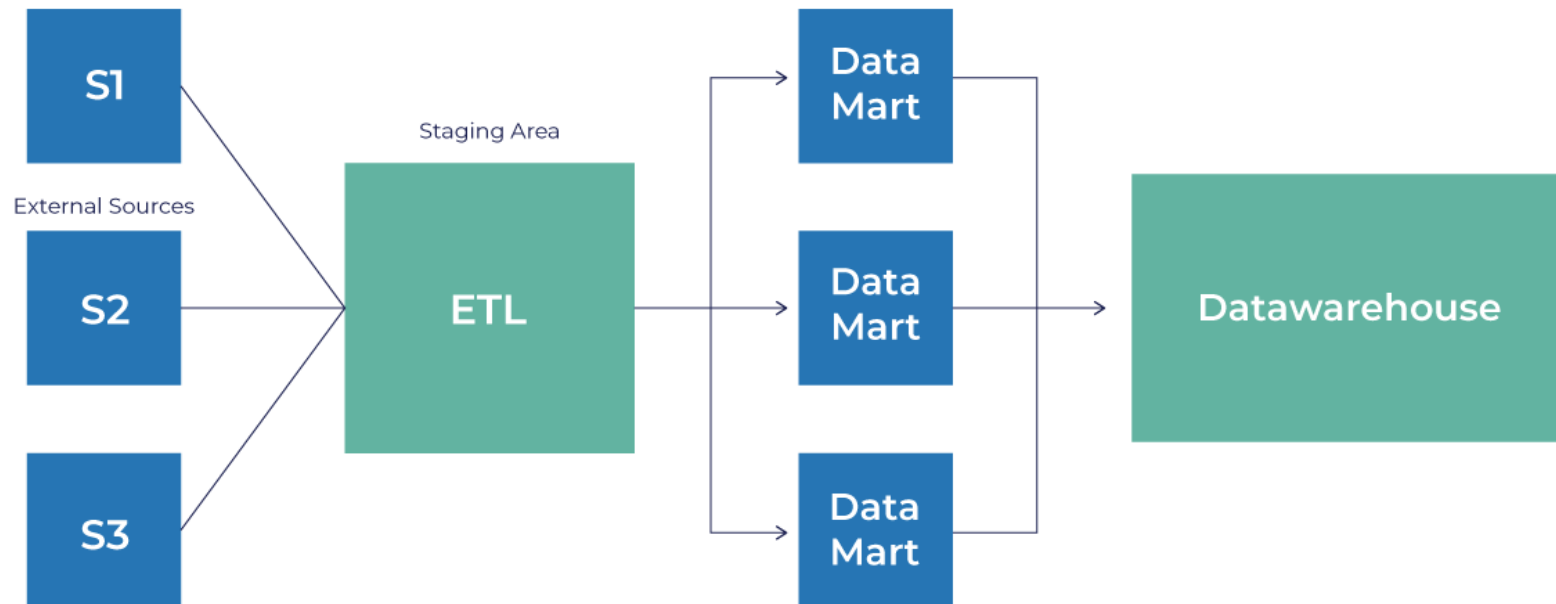
DW Design



DW Design – top-down



DW Design – bottom-up



DW Design – hybrid

The hybrid methodology combines elements of both Inmon and Kimball approaches, aiming to balance enterprise integration with rapid delivery.

Characteristics:

May start with a core, normalized data warehouse for critical subject areas, while simultaneously developing dimensional data marts for high-priority business needs.

Uses operational data stores (ODS) and data vaults to handle integration and flexibility.

Supports both detailed atomic data and summarized, business-focused data marts.

Advantages:

Balances speed and integration.

Allows organizations to adapt to changing requirements.

Reduces risk by delivering value early while building for the long term.

Disadvantages:

Can increase architectural complexity.

Requires careful governance to avoid data inconsistencies.

Approach

Aspect	Supply-Driven Approach	Demand-Driven Approach
Starting Point	Data source analysis	User/business requirements analysis
User Involvement	Secondary, mainly for data selection	Primary, drives the entire design process
ETL Complexity	Simplified, direct mapping from sources	More complex, may require data transformation
Risk	DW may not meet user needs	Requirements may be hard to specify/match
Best Used When	Data sources are well-known and structured	Business goals are dynamic, requirements unclear
Flexibility	Less flexible to changing business needs	More adaptable to evolving requirements

Approach

Goal-driven approach

Business requirements

„Dynamic business growth“

„Dynamic business growth“

What is this growth?

How is it understood?

Is it the expansion of our product offer?

Is it the elimination of non-profitable branches?

Is it the expansion on new territories?

Is it focusing on a particular target group?

Provides hint

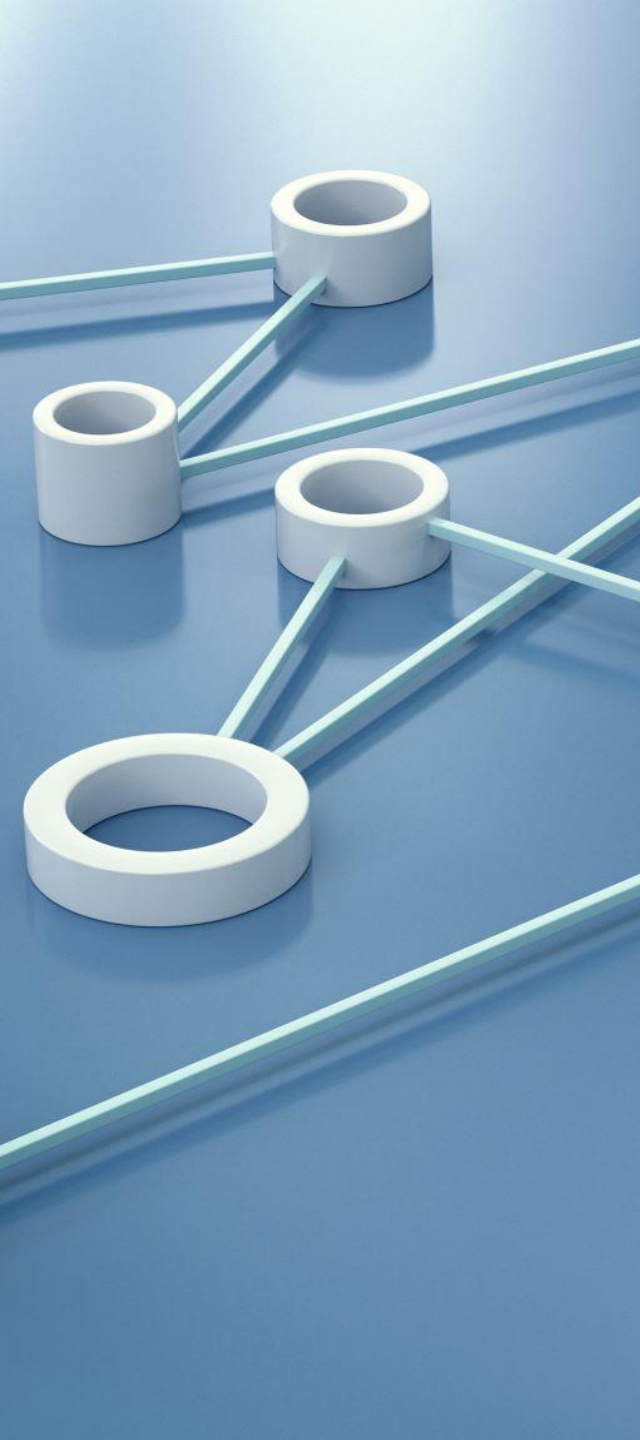
The major factors affecting

the design of the data warehouse

reflect its primary mission, which is to serve as a collection point for the needed data stored in various operational systems and as a distribution point for sending this data to the data marts.

the content of the data warehouse

are the information needs of the people who will use the resultant data marts and the organization of the data in the source systems.



DW Design

Key elements to consider:

- Understand the domain and business goals.

- Identify relevant data sources.

- Define the data destination schema.

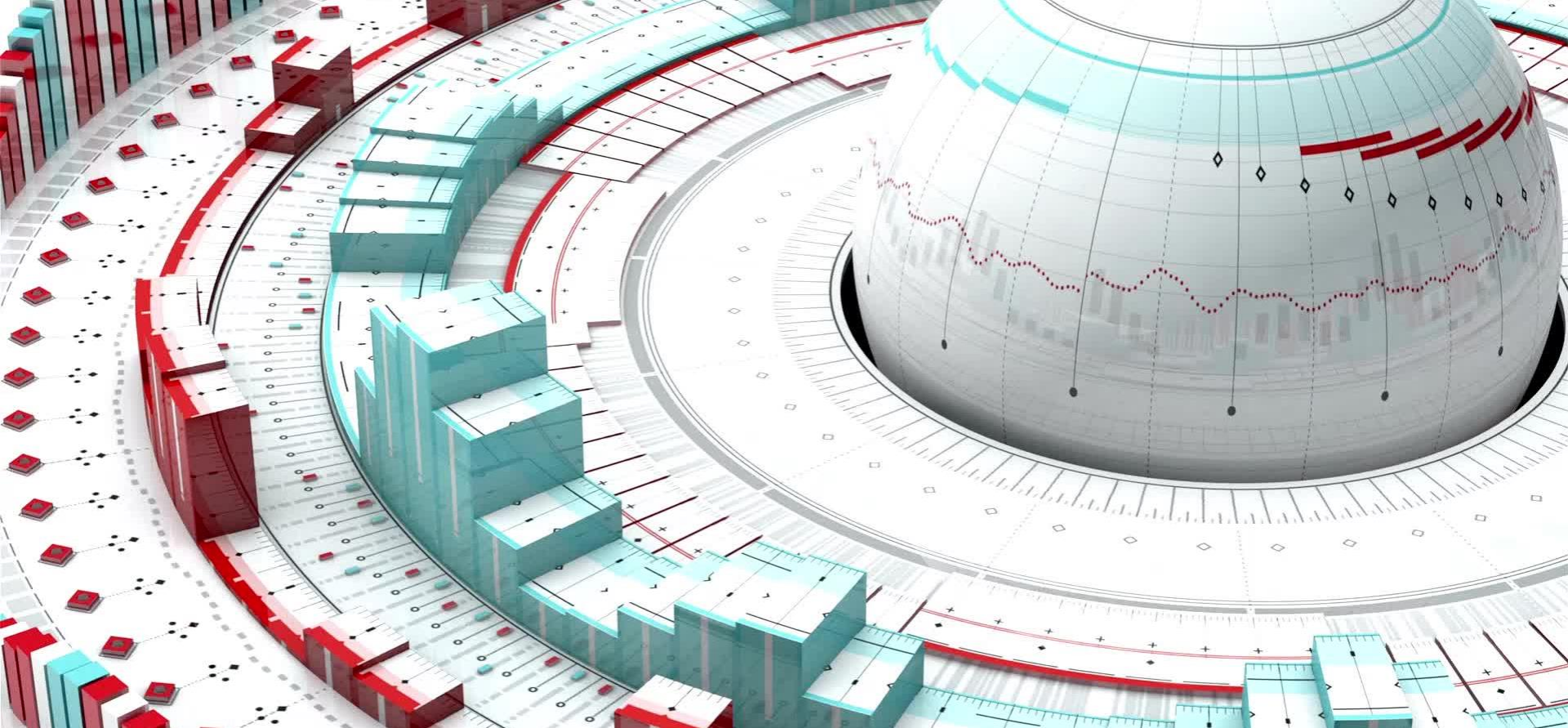
- Create the data warehouse design schema.

- Plan the ETL process for the data warehouse design.

- Choose appropriate hardware, software, and tools.

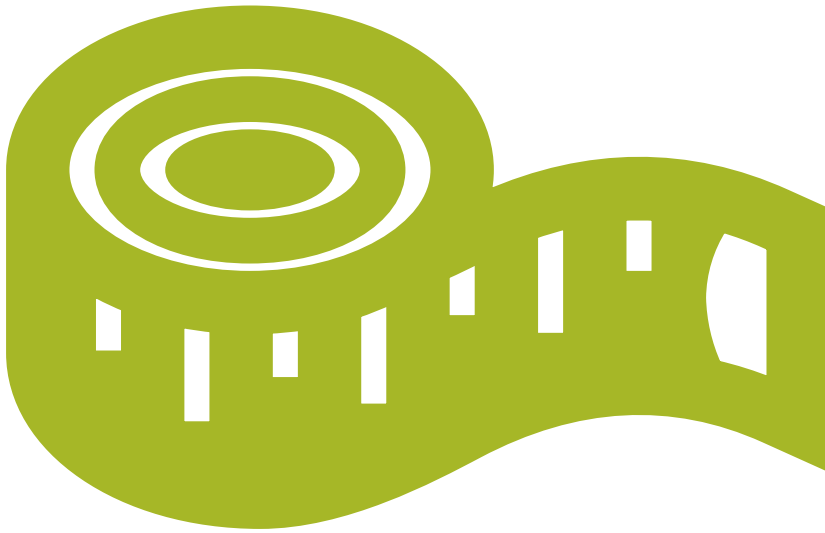
- Deploy and maintain the warehouse.

- Managing data quality and performance over time.



DW Dimensional Design

Facts



Facts

A fact is an element of the multi-dimensional space

Associates a set of dimension elements with measures

„cube cell“

Granularity relates to dimensions

Fact is uniquely identified through a combination of dimension elements

Granularity should be defined – depending on user requirements and data availability

Single granularity for a fact table

Qualifying and quantifying information

Dimensions



Dimensions

Perspective of an application area

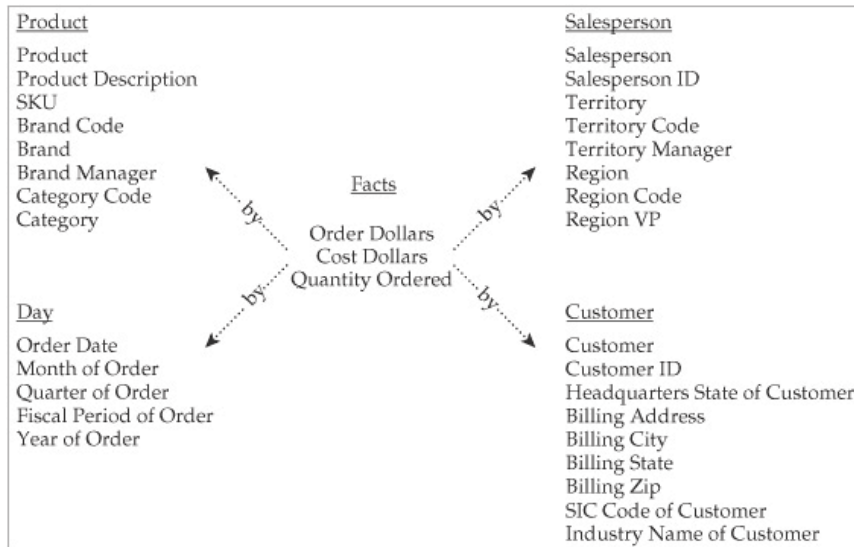
Contain the **entry points** and **descriptive labels** that enable the DW/BI system to be leveraged for business analysis

Qualifying information

Consists of multiple attributes

descriptive attributes used by BI applications for filtering and grouping the facts

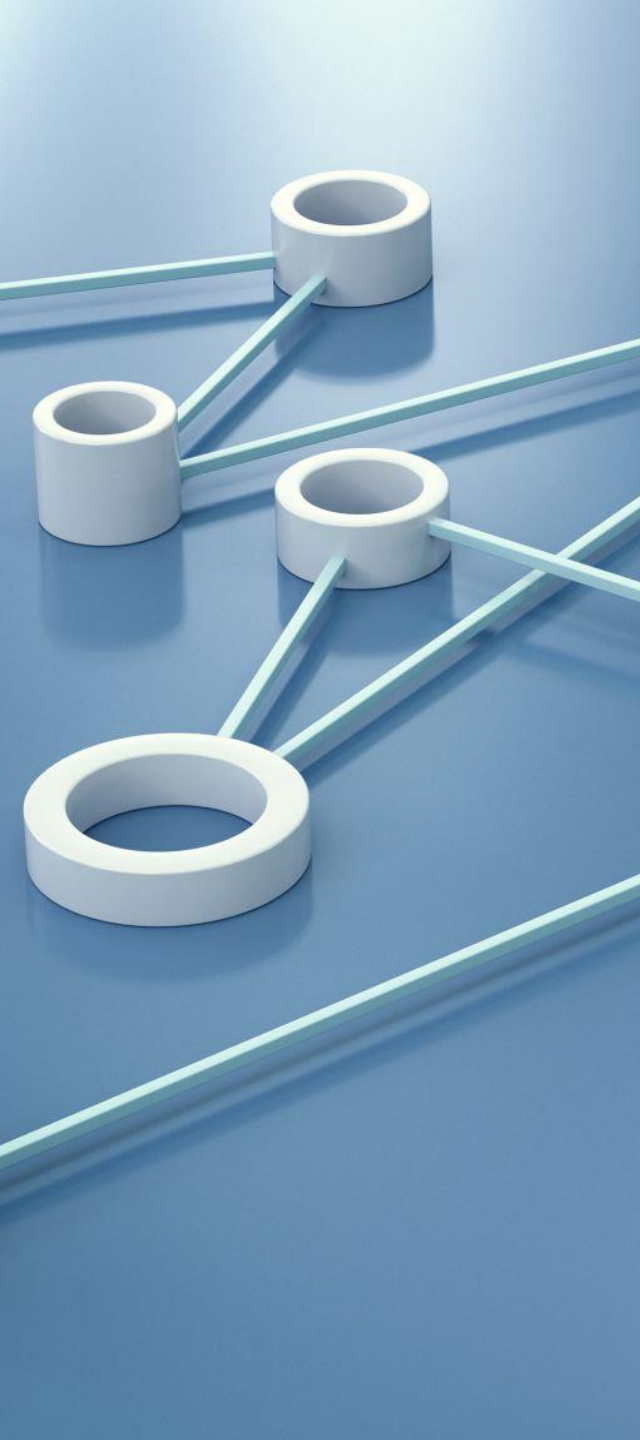
Stars



The diagram can be easily understood by a businessperson.

“Things I measure” go in the middle;

“the ways I look at them” go in the surrounding boxes, loosely grouped or categorized.



DW Design

Requirements analysis

gathering knowledge about business requirements in the field of analytical processing

Logical design of a data warehouse

conceptual definition of required data structures

ETL software implementation

Design of software modules used to power data warehouses

Implementation of analytical applications

Implementation of programs for end users

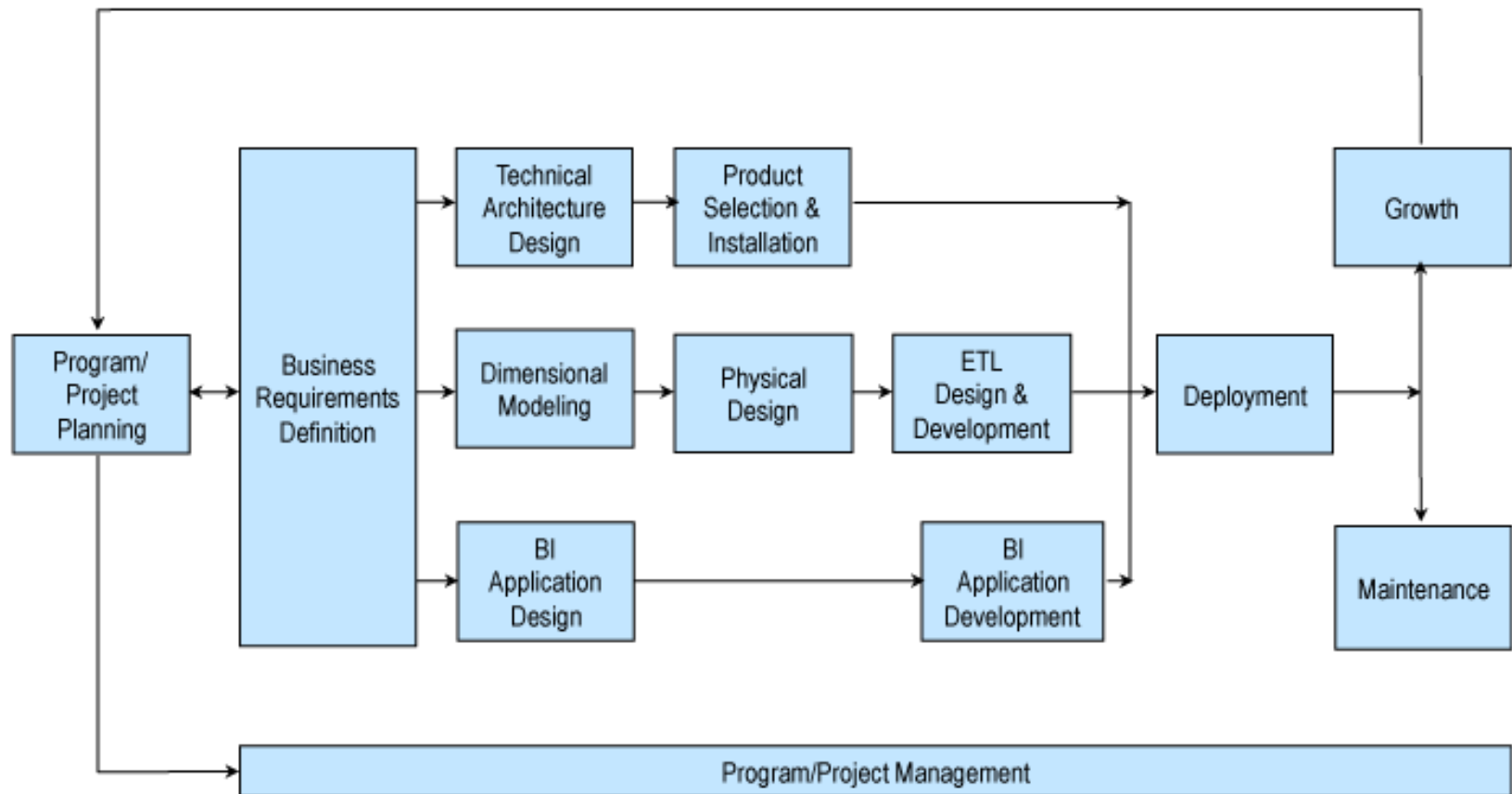
Tuning a data warehouse

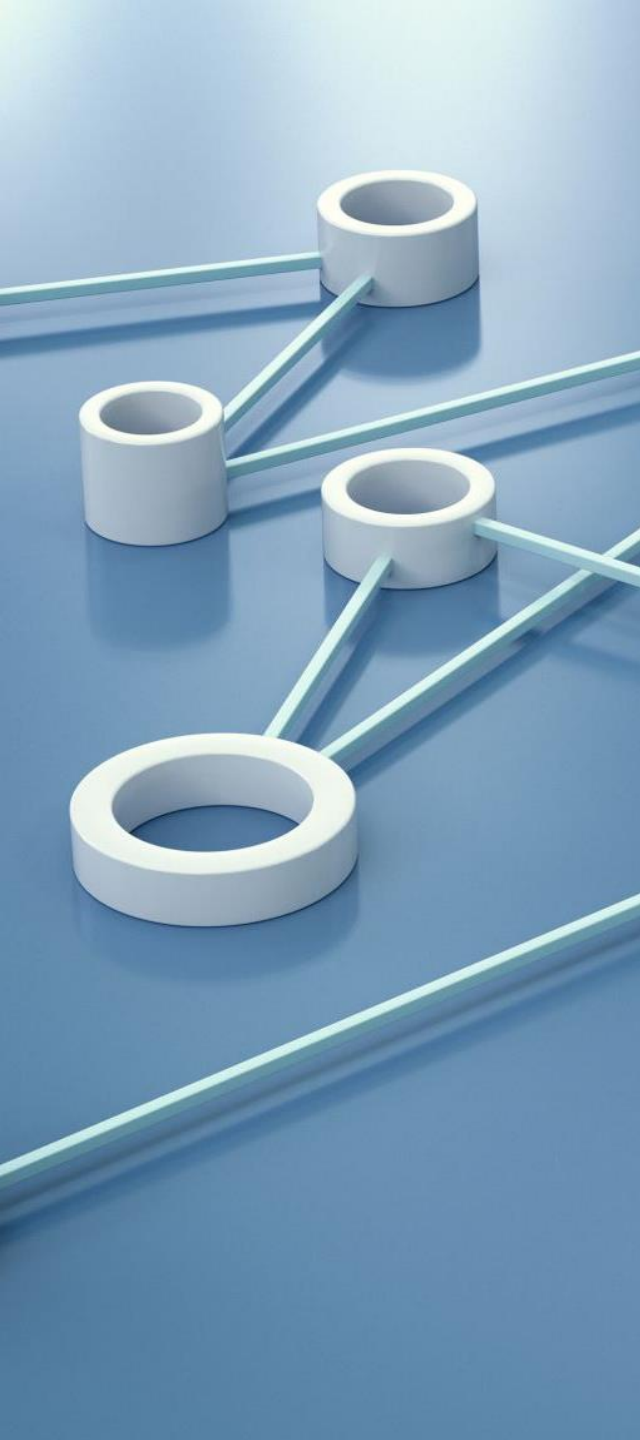
Reconfigure the database server, create additional indexes and materialized perspectives

Refresh a data warehouse

Development and implementation of the data upgrade process in the data warehouse

DW Design





DW Design

The requirements need to be in sufficient detail to move on to steps of designing the data model, data integration processes, and BI applications.

A complete set of requirements would include the following subjects:

Business requirements

High-level business requirements, Business processes supported, Business rules and metrics

BI functional requirements

Use cases, Process workflow and user interaction, Analytical styles and functionality

Data requirements

Data sources; Data conformance, consistency, and currency; Data integration; Data quality

Regulatory and compliance requirements

Country, Industry, Privacy and security

Technical requirements

Infrastructure standards, Technology directions



Basics of Dimensional Design

Dimensional Design

We sell products in various markets, and we measure our performance over time

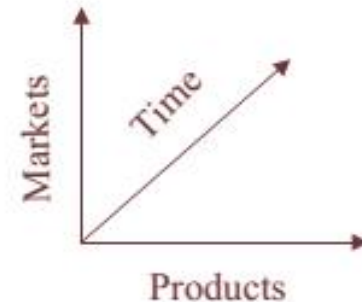


Business Manager



We sell *Products* in various *Markets*, and we measure our performance over *Time*

Data Warehouse Designer



Dimensional and ER

ER Model

OLTP systems

Data consistency, non-redundancy, and efficient data storage are critical

Entity-Relationship Modelling

Removes data redundancy

Ensures data consistency

Expresses microscopic relationships

Dimensional Model

Focus is on how managers view the business

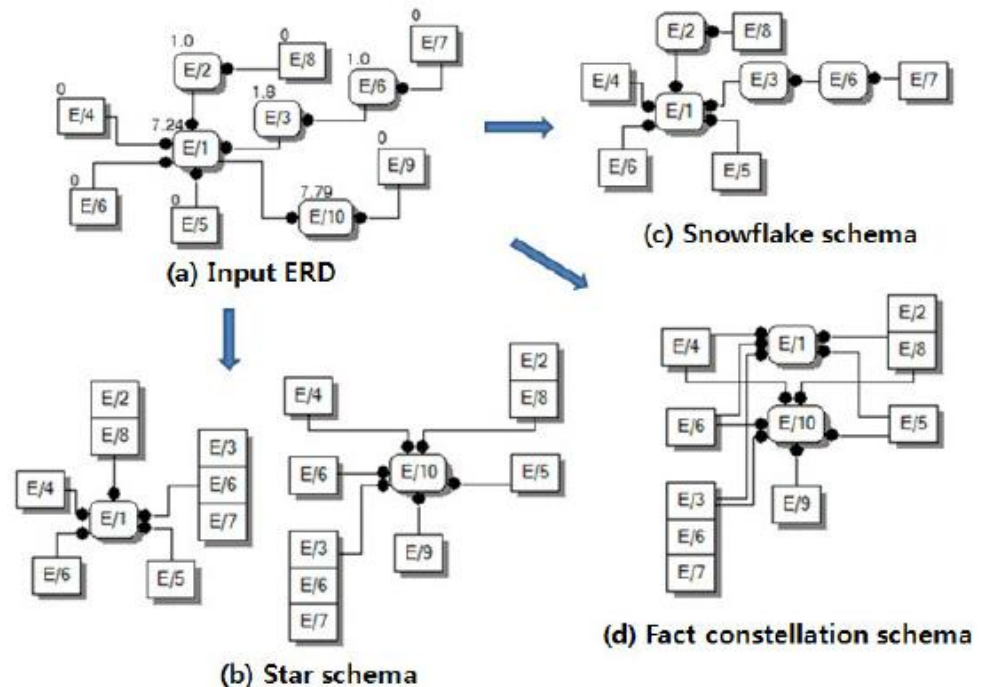
Information is centered around a business process – how the business measures the process in many ways along several business dimensions

Dimensional Modelling

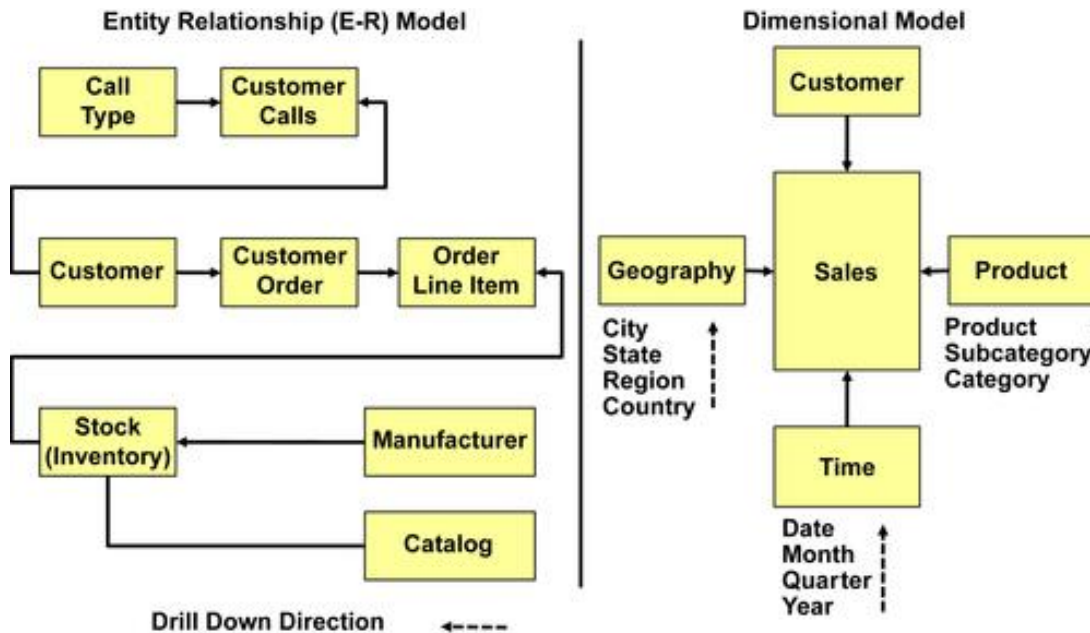
Captures critical measures

Views along dimensions

Intuitive to business users



Dimensional and ER



Why do these structures matter for data modeling?

Each structure has its purpose, and you need the right structure for the job at hand.

The choice is highly dependent on the way the data is used and the type of data warehousing or business intelligence tool.

Dimensional Design

Information about sales

What is meant by the term sale?

A sale can occur when

the order has been received from a customer,

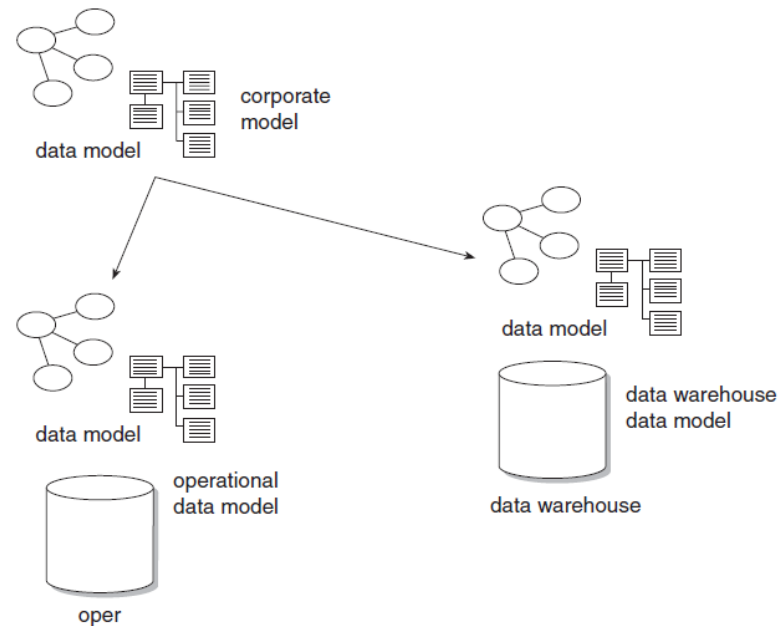
the order is sent to the customer,

the invoice has been raised against the order

Need to understand

needs of the business

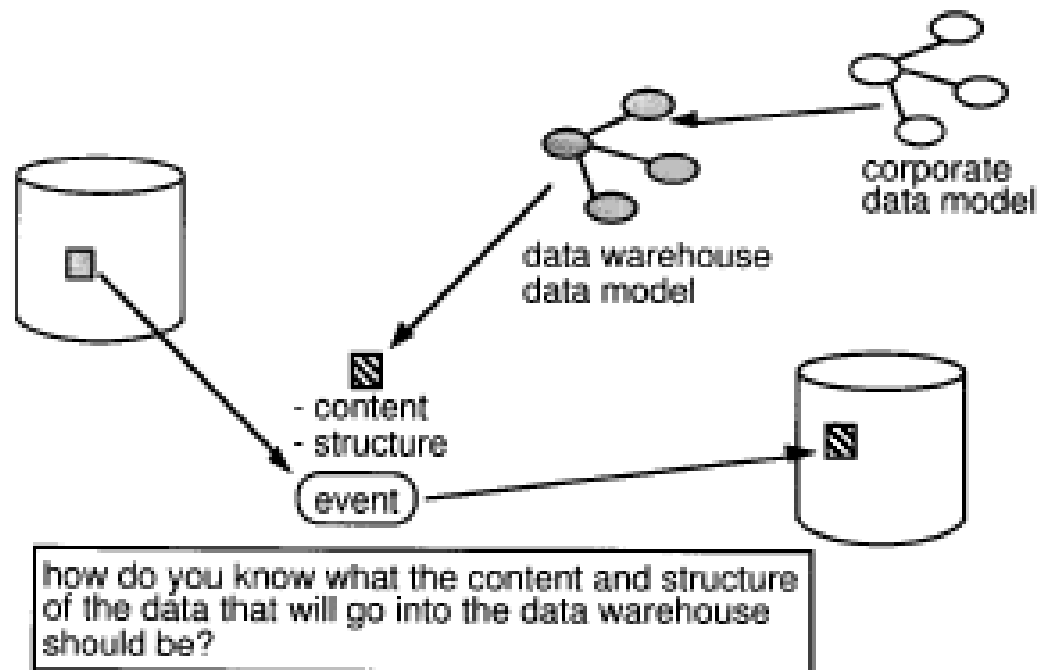
realities of the underlying source data



- operational data model equals corporate data model
- performance factors are added prior to database design

- remove pure operational data
- add element of time to key
- add derived data where appropriate
- create artifacts of relationships

Dimensional Modelling



Dimensional Design

Starting point should be the corporate data model

Changes are made to the corporate data model as it is applied to the data warehouse

- Remove pure operational data

- Add element of time to key

- Add derived data where appropriate

- Create artifacts of relationships

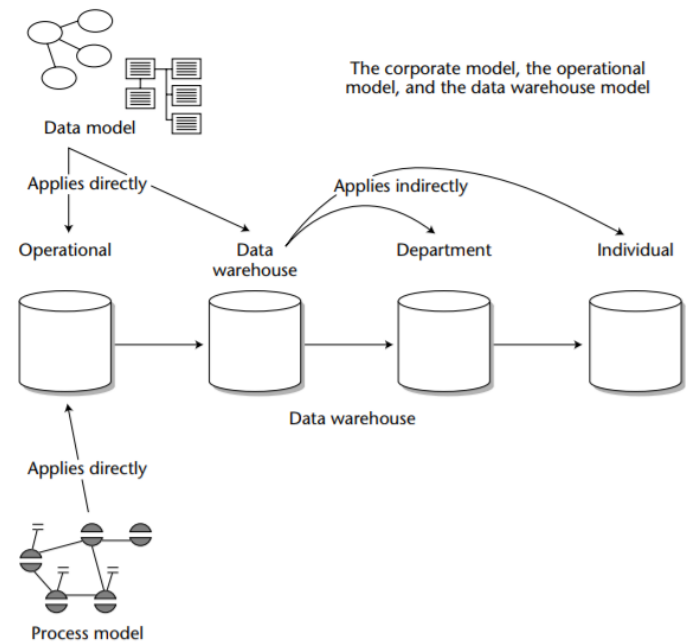


Figure 3-7 How the different types of models apply to the architected environment.

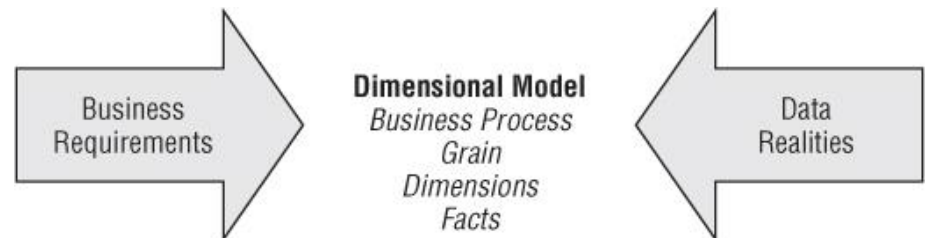
Dimensional Modelling

You need to consider both your business users' requirements and the realities of your source data in tandem to make decisions

Resist the temptation to model the data by looking at source data alone.

It may be less intimidating to dive into the data rather than interview a business person

the data is no substitute for business user input.



MAKE	MODEL NAME	TARGET GROUP	DESCRIPTION
Zenith	Zipster	The young at heart (and age)	The Zipster is a sporty, subcompact-class car with a small price tag, excellent gas mileage, and limited options. This is the low-end offering in the Zenith line of cars.
Zenith	Zombie	Older, retired drivers with a limited income	The Zombie is a compact-sized, four-door automobile, noted for its economical upkeep and good gas mileage.
Zenith	Zoo	Families with small children	The Zoo is a four-door, mid-size car. The car is moderately priced and has good gas mileage.
Zenith	Zoom	Sports car enthusiast of modest means seeking excitement	The Zoom is a moderately expensive, big-engine performance car that offers quick response, agile handling, and fast acceleration.
Zenith	Zeppelin	Luxury minded individual	The Zeppelin is the top-of-the-line Zenith car offering unsurpassed quality and features. It is a four door, full-sized model.
Tuxedo	Topsail	Young professionals	The Topsail is a mid-sized, two-door sedan equipped with a full complement of luxury features, including leather seats, an eight-way power-adjustable seat, a tilt steering wheel, and a high-tech car alarm.
Tuxedo	Tiara	The truly discriminating, sophisticated driver	The Tiara is a full-sized four-door sedan that is the top of the line Tuxedo automobile and is priced accordingly. It has many of the same features found in the Topsail but offers every conceivable luxury, including seat and outside mirror heaters.
Tuxedo	Thunderbolt	Wealthy sports car enthusiasts	The Thunderbolt marks an acknowledged milestone in sports cars. It combines all the breathtaking performance of a thoroughbred with the ease of operation, comfort, and reliability of a passenger car.

Dimensional Modelling

Zenith Automobile Company (ZAC)

automotive manufacturing firm

manufactures two makes of automobiles:

Zeniths

higher-end luxury Tuxedos

cars are sold through dealers throughout the United States

dealers are located within sales areas, which are grouped into sales territories, which are grouped into sales regions

SUBJECT AREA	DEFINITION	EXAMPLES	REMARKS
Sales	Transactions that shift the ownership or control of a product from the Company to a Customer	•Sales Transaction •Sales Transaction Detail •Credit Memo	Sales is actually an associative subject area in that it is the intersection of the Customer, Store, Product, and so on. Some companies may choose to include the entities related to sales within one of those subject areas instead.
Suppliers ¹	Legal entities that provide the company with goods and services	•Broker •Manufacturer •Supplier	In the case of a contractor, the person doing the work is included in Human Resources, and the company that provides that person is included in Suppliers.
Locations	Geographical points or areas	•Geopolitical Boundary •Country •Address	This can be expanded to include electronic locations such as email addresses and phone numbers.
Materials	Goods and services that are used or consumed by the company or that are included in a piece of equipment, facility, or product	•Chemical •Fuel •Supply	Sometimes, a product is used as a component of another product. When this is the case, a relationship between the relevant entities will be indicated in the business data model.
Products	Goods and related services that the company or its competitors provide or make available to Customers	•Product •Service •Advice	Competitor items that the company does not provide are often included to facilitate monitoring these and to support future decisions.

Dimensional Modelling

A fully developed business data model may contain hundreds of entities.

A subject area model, which defines the major groupings of information, is a good way to manage these entities by providing a logical approach for grouping the entities.

Subject area model

Only the subject areas needed to answer the business questions:

Automobiles are the vehicles and associated parts manufactured by Zenith Automobile Company and sold through its dealers.

Customers are the parties that acquire automobiles and associated parts from Dealers.

Dealers are agencies authorized to sell Zenith Automobile Company automobiles and associated parts.

Factories are the facilities in which Zenith Automobile Company manufactures its automobiles and parts.

Incentive Programs are financial considerations designed to foster the sale of automobiles.

Sales Organizations are the groupings of Dealers for which information is of interest.

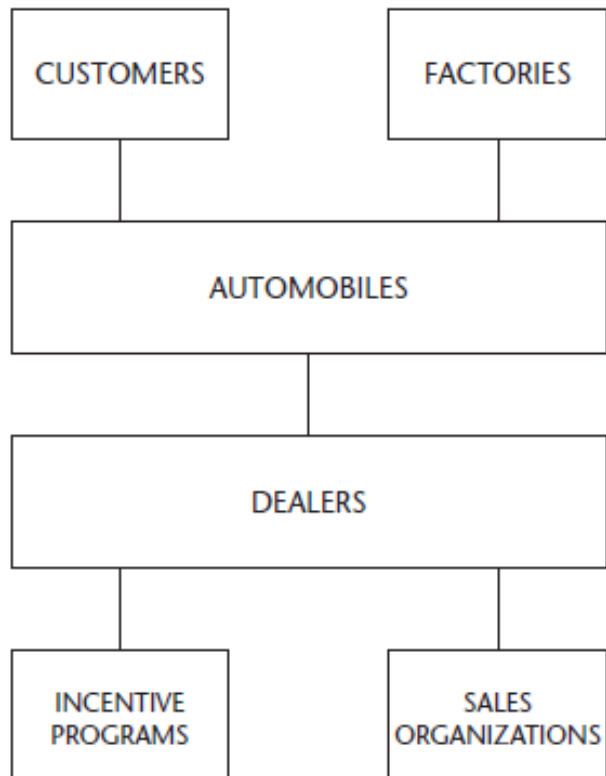


Figure 3.5 Zenith Automobile Company partial subject area.

Based on interviews with key stakeholders:

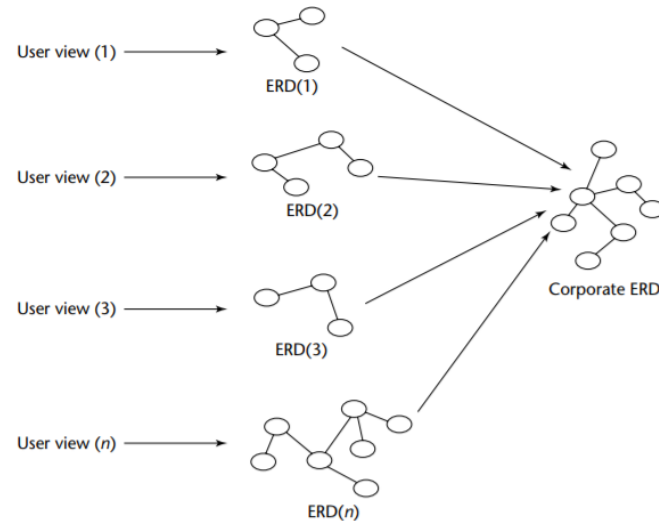
- What is the monthly sales trend in terms of quantity and dollar amounts sold of each make, model, series, and color (MMSC) for a specific dealer, by each sales area, sales territory, and sales region, for each state and for each metropolitan statistical area (MSA)?
- What is the pattern in the monthly quantity of inventory by MMSC for each dealer, by each sales area, sales territory, sales region, and MSA?
- How does the monthly quantity and dollars of sold automobiles by MMSC having a particular emissions type—by Dealer, Factory, Sales Area, Sales Territory, and Sales Region—compare with the same time frame last year and the year before?
- What is the trend in monthly actual sales (dollars and quantities) of MMSC for each dealer, sales area, sales territory, and sales region compared to their objectives? Users require this information both by monthly totals and cumulative year to date (YTD).
- What is the history (two-year comparisons) of the monthly quantity of units sold by MMSC and associated dollar amounts by retail versus wholesale dealers?
- What are the monthly dollar sales and quantities by MMSC this year to date as compared to the same time last year for each dealer?
- What is the monthly trend in sales dollars and quantities by MMSC for particular types of incentive programs, by dealer, sales area, sales territory, sales region, and MSA?
- What is the monthly trend in the average time it takes a dealer to sell a particular MMSC (called velocity and equal to the number of days from when a dealer receives the car to the date it is sold) by sales area, sales territory, sales region, and MSA?
- What was the monthly average selling price of an MMSC for each dealer, sales area, sales territory, sales region, and MSA?
- How many days was a dealer placed on credit hold for this month only and for the entire year? In addition, what was the total number of months in the past two years that the dealer was put on credit hold?
- Compare monthly sales dollars and quantities from the last body style (body style is make + model) to the current body style for each sales region? Body styles change every four years.

Dimensional Modelling

Dimensional Design

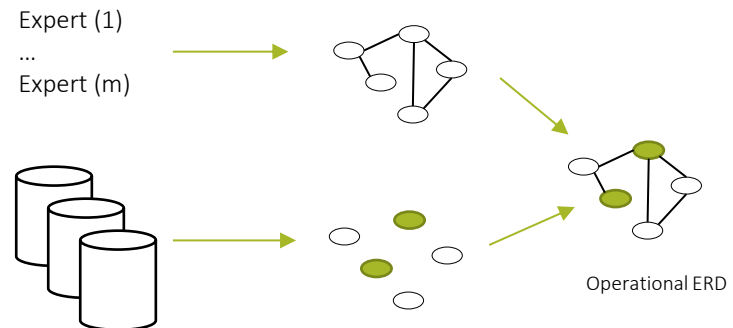
The high level of modelling representing the known requirements are created by means of user view sessions

which are interview sessions with the appropriate personnel in the various departments



Data realities are uncovered by meeting with source system experts

doing high-level data profiling to assess data feasibilities



Dimensional Design

There are three major approaches:

Closed room

modelers working on their own with little or no involvement by business representatives

Interviews

provide an excellent means of obtaining information from individual business representatives

Facilitated sessions

sessions involve representatives of the various business areas, just as the interviews do

significant difference is that the people are interacting with each other instead of providing individual contributions

Dimensional Design

Figure 4.3 Reducing the entities to be considered.

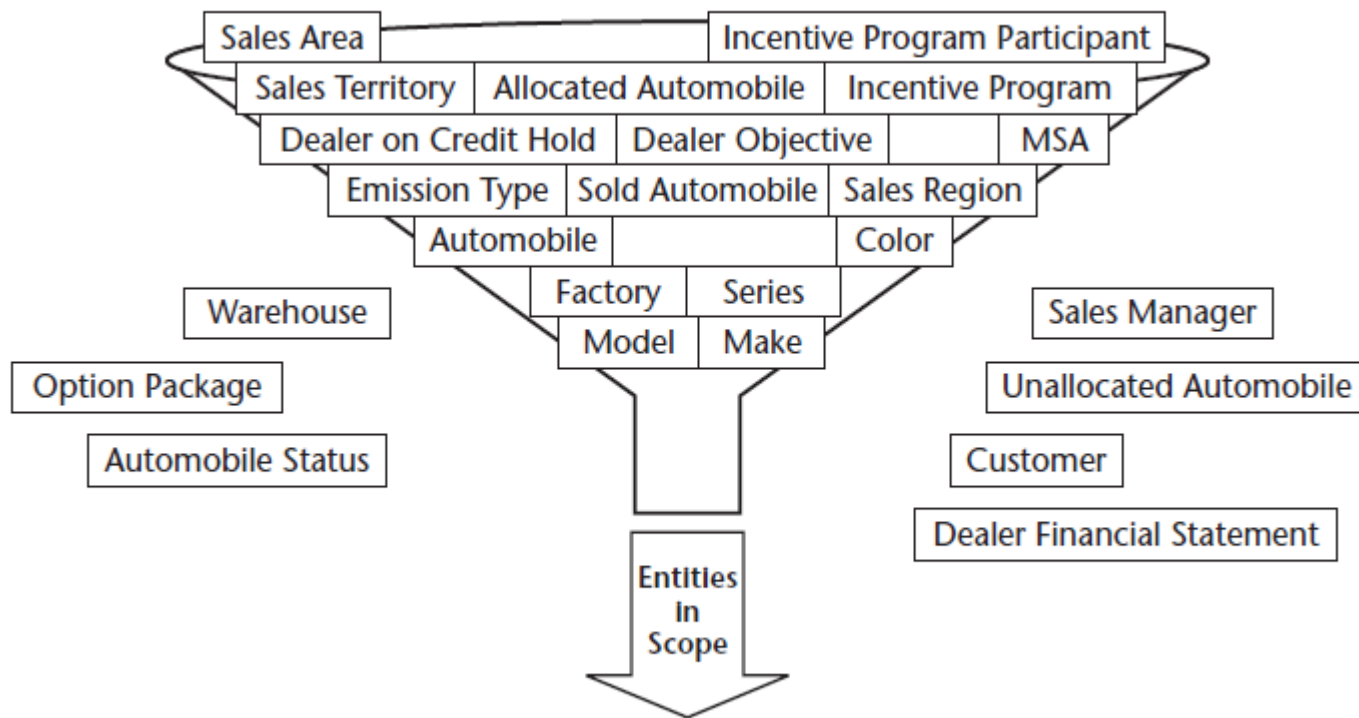


Figure 4.3 Reducing the entities to be considered.

Dimensional Design

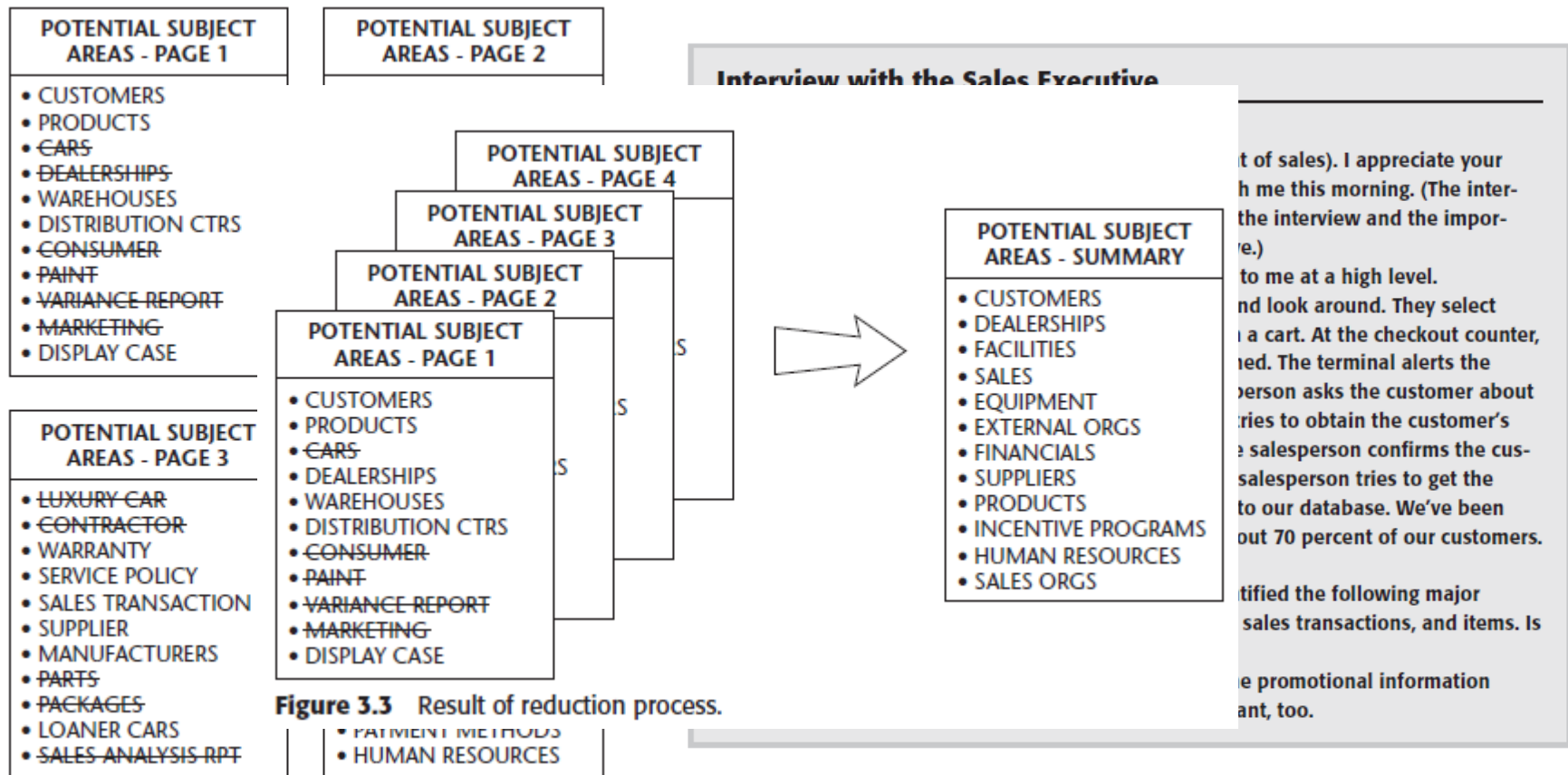


Figure 3.3 Result of reduction process.

Figure 3.2 Result of refinement process.

Dimensional Design

Requirements:

- General overview of the company

 - it's goals are reporting programs and ad hoc queries that access data that is integrated from multiple data sources*

 - Do not focus on transactional processing – DW is all about querying and reporting*

Interviews

 - High-level, general information from CEOs and management*

 - Detailed needs from workers (financial, marketing, planning and development)*

 - Exemplar questions

Existing reports, solutions

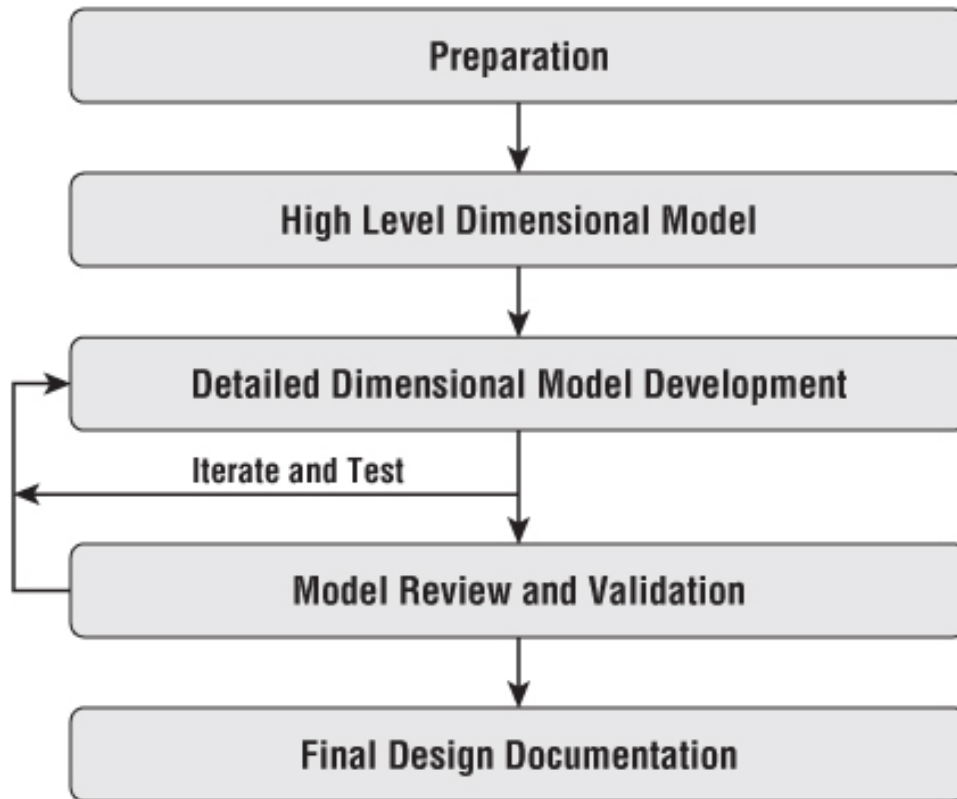
 - Trap - repeating same mistakes*

 - Trap - creating just a better version of the existing solution*

 - Trap - moving operational reports to DW*

This should allow you to design a conceptual model

Dimensional Design



Inputs:

Business Data Model

Scope Document

Information Requirements

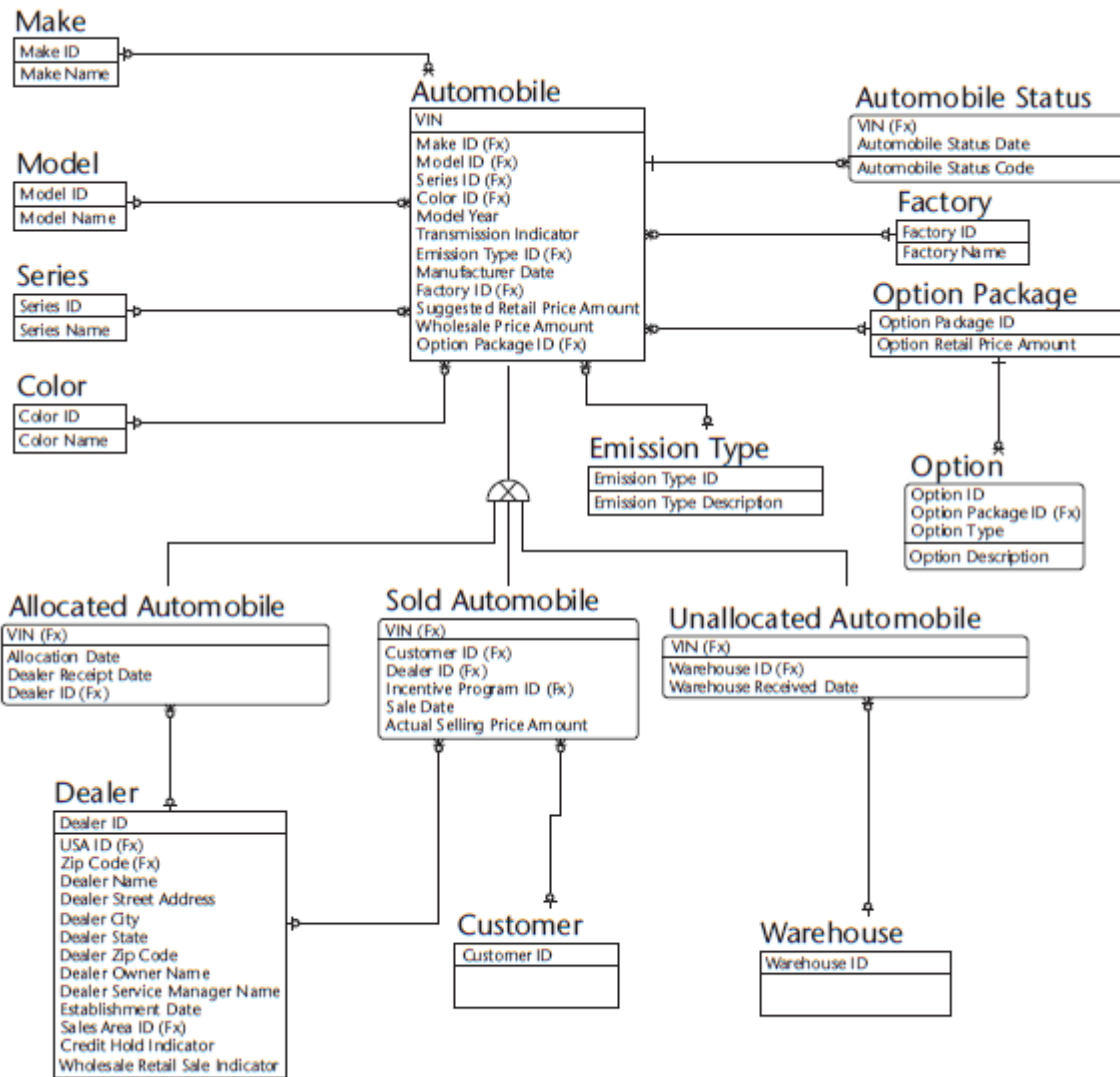
Existing Reports and Queries

Source Data

Prototype

Figure 3.8 Entity-relationship diagram—entities.

Figure 3.9 Entity-relationship diagram—entities and attributes.



Selection

The selection of the data elements for actual inclusion is not a simple process.

Consider a business question:

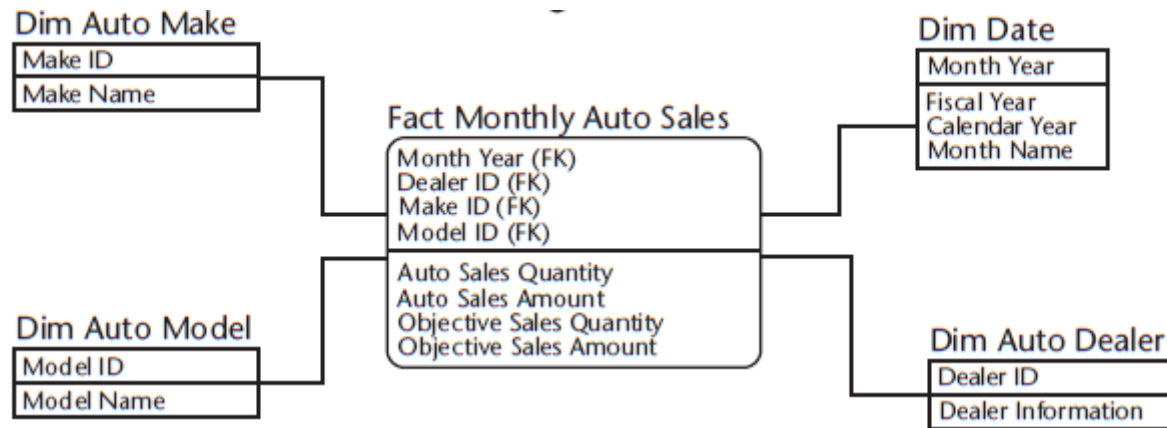
“What is the monthly sales trend in quantity and dollar amounts sold for each Make, Model, Series, and Color (MMSC) for a specific dealer, by each Sales Area, Sales Territory, and Sales Region, for each state and for each Metropolitan Statistical Area (MSA)?”

Three groups:

- definitely needed (date, sales quantity, sales amount)

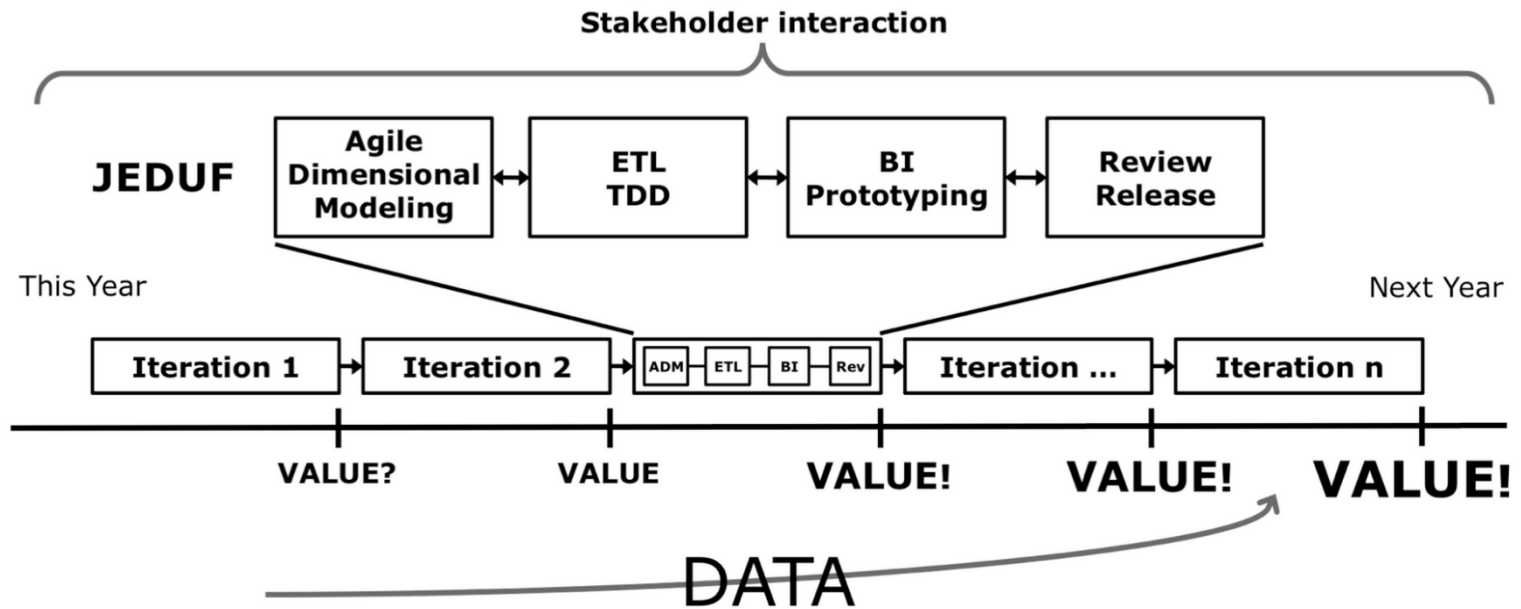
- definitely not needed (record update date and time)

- might be needed (time of sale, special discount, sales tax amount)



Data mart of users
interested in monthly sales
and objectives tracking
remains unchanged

Agile Dimensional Design





Dimensional modelling

Dimensional Model Design Life Cycle – DMDDL

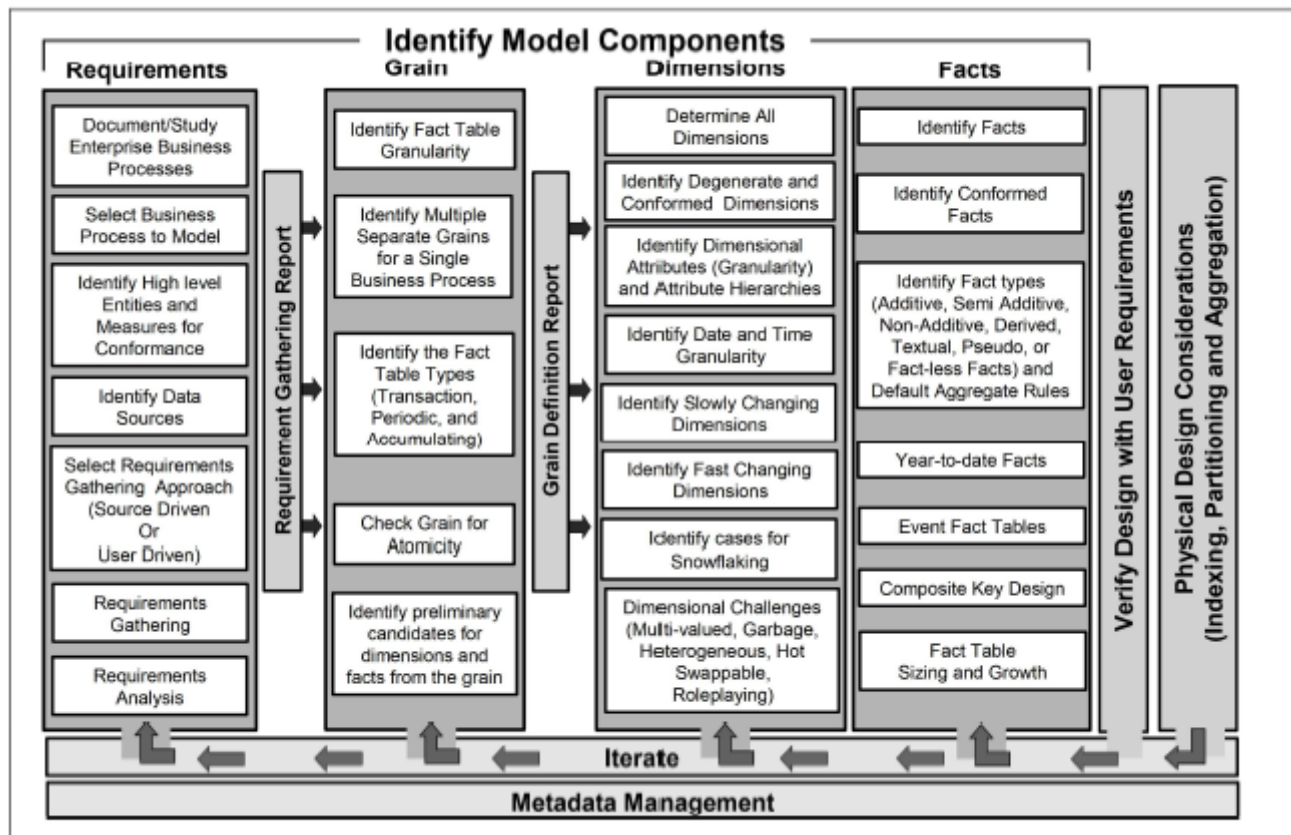


Figure 5-1 Dimensional Model Design Life Cycle

Dimensional Modelling

Four Steps in Dimensional Modelling

Choose the Business Process

Select the business process to model

(e.g. sales).

Choose the Grain

Determine the grain of the business process

(e.g. single transaction in a market identified by bar-code scanners at cash register)

Choose the Dimension

Choose the dimensions describing the business process

(e.g. localization, product, data, promotion, etc.).

Define the measures

Identify the numeric measures for the facts

(e.g. price, quantity).

Dimensional Modelling

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Establish Relationships

Link fact and dimension tables using one-to-many relationships

Optimize Granularity

Ensure fact tables consistently store data at the lowest required detail level

Dimensional Modelling

Business processes

are the operational activities performed by your organization

in dimensional modeling refers to a set of related operational activities or events that generate or capture measurable data-such

such as taking an order, processing an insurance claim, registering students for a class, or snapshotting every account each month.

Business process events generate or capture performance metrics that translate into facts in a fact table.

Why This Step Matters

Foundation for Modeling

determines the fact table's focus and the relevant dimensions, setting the stage for all subsequent modeling decisions

Avoids Redundancy

modeling around business processes rather than departments prevents data duplication and maintain consistency across the warehouse

Supports Strategic Analysis

ensures the data warehouse is aligned with organizational objectives and supports high-value analytical queries

Dimensional Modelling

Choose business process

Goal

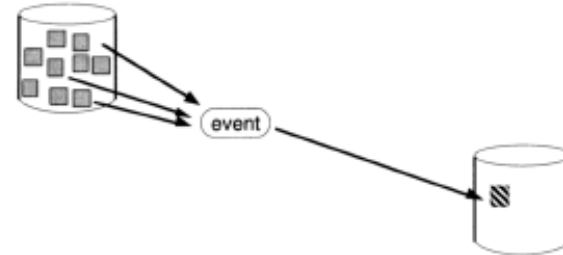
Define the purpose, and decide on the subject(s) for the data warehouse

Snapshots

the snapshot has been taken as a result of an event

The first step in designing the data warehouse is to identify the events

Business process events generate or capture performance metrics that translate into facts in a fact table.



Dimensional Modelling

Snapshots

EVENTS

The most basic consideration of a snapshot is that the snapshot has been taken as a result of an event.

The event may be triggered by a wide variety of occurrences:

- an occurrence of a transaction,
- the periodic passage of time,
- a threshold having been reached,
- an audit,
- a special request, etc.

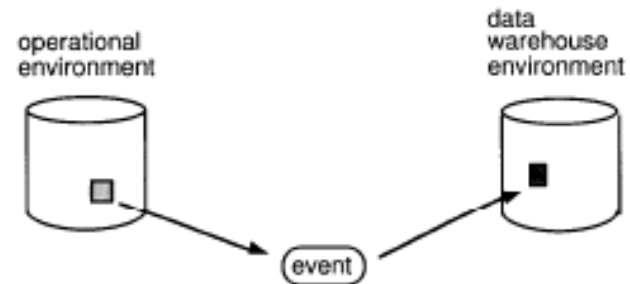


Figure 2: Data is moved into the data warehouse as a result of the occurrence of some event.

Dimensional Modelling

Select the Business Process – define facts

To identify your organization's business processes, it's helpful to understand several common characteristics:

Business processes are frequently expressed as action verbs because they represent activities that the business performs.

The companion dimensions describe descriptive context associated with each business process event.

Business processes are typically supported by an operational system
such as the billing or purchasing system.

Business processes generate or capture key performance metrics.

Sometimes the metrics are a direct result of the business process; the measurements are derivations at other times.

Analysts invariably want to scrutinize and evaluate these metrics by a seemingly limitless combination of filters and constraints.

Business processes are usually triggered by an input and result in output metrics.

In many organizations, there's a series of processes in which the outputs from one process become the inputs to the next.

Dimensional Modelling

Typically

you'll be able to identify multiple
business processes

Focus on

high impact

Importance

low risk

Data quality, complexity, access, etc.

The goal is to select the business
process that is most critical to the
organization's analytical needs

Business Process



Dimensional Modelling

Choose the grain

The grain establishes exactly what a single fact table row represents

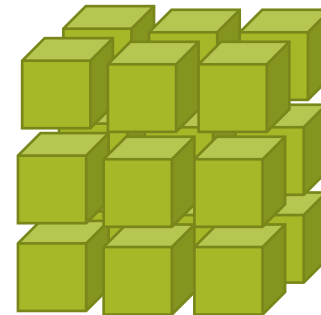
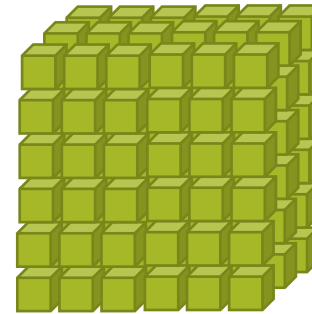
Declaring the grain is the pivotal step in a dimensional design

declaration becomes a binding contract on the design

The grain must be declared before choosing dimensions or facts

every candidate dimension or fact must be consistent with the grain

consistency enforces a uniformity on all dimensional designs that is critical to BI application performance and ease of use



Dimensional Modelling

Goal

Grain declarations are expressed in business terms.

Although the grain ultimately is equivalent to the primary key, it's a mistake to list a set of dimensions and then assume this list is the grain declaration.

Whenever possible, you should express the grain in business terms.

Declaring the grain is a critical step that can't be taken lightly.

Dimensional Modelling

Declare the Grain

It provides the answer to the question, “How do you describe a single row in the fact table?” The grain is determined by the physical realities of the operational system that captures the business process's events.

Example grain declarations include:

- One row per scan of an individual product on a customer's sales transaction

- One row per line item on a bill from a doctor

- One row per individual boarding pass scanned at an airport gate

- One row per daily snapshot of the inventory levels for each item in a warehouse

- One row per bank account each month

Dimensional Modelling

Goal

Define the fine grain level of data

Identify the lowest possible level of detail – atomic grain

Atomic grain refers to the lowest level at which data is captured by a given business process

Start by focusing on atomic-grained data

because it withstands the assault of unpredictable user queries

rolled-up summary grains are important for performance tuning

but they pre-suppose business's common questions

each proposed fact table grain results in a different physical table

different grains must not be mixed in the same fact table

Dimensional Modelling

Identify the Dimensions

Dimensions fall out of the question

“How do business people describe the data resulting from the business process measurement events?”

You need to decorate fact tables with a robust set of dimensions representing all possible descriptions that take on single values in the context of each measurement.

If you are clear about the grain, the dimensions typically can easily be identified as they represent the “who, what, where, when, why, and how” associated with the event.

Examples of common dimensions include

date, product, customer, employee, and facility.

Dimensional Modelling

Identify the Dimensions

With the grain of a fact table firmly in mind, all the possible dimensions can be identified.

In a spoken or written statement, the word “by” is almost always followed by a dimension.

“What are order dollars by product category for January?”

the word “for” is also a good indicator of the presence of a dimension.

“What are order dollars by product category for January?”

With the choice of each dimension, you then list all the discrete, text-like attributes that flesh out each dimension table.

Dimensional Modelling

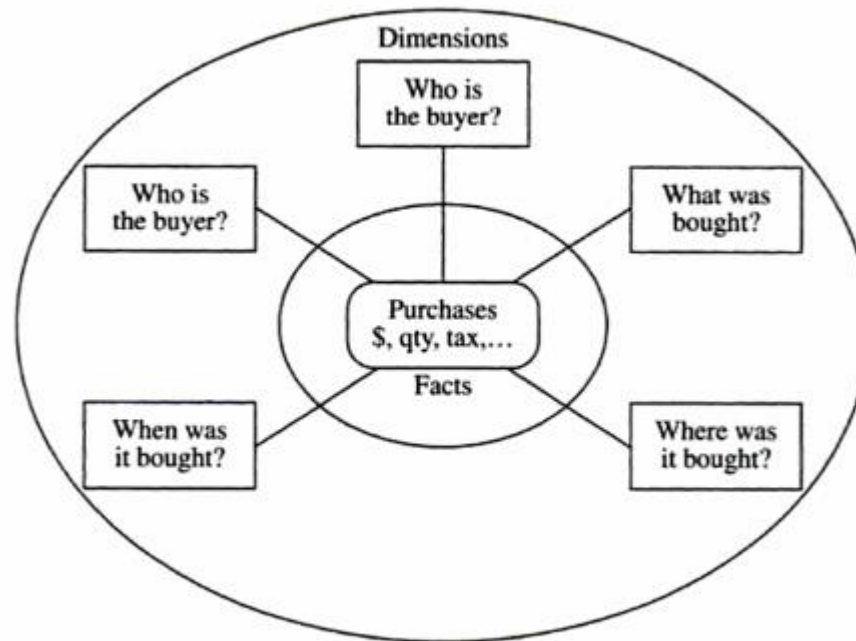


Exhibit 14.3. The star model.

Dimensional Modelling

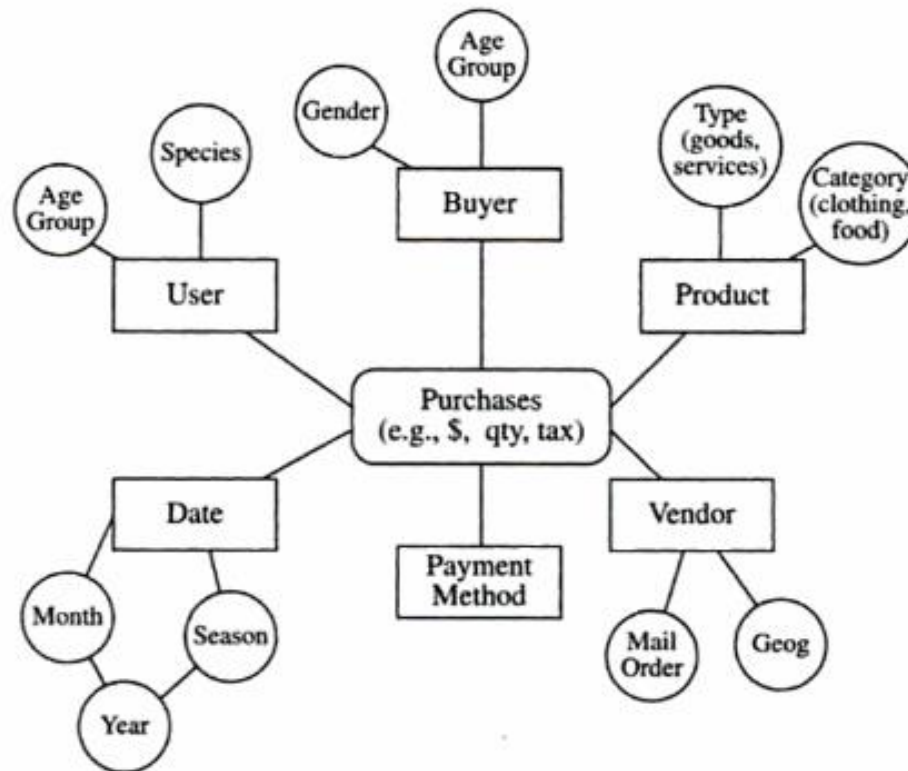


Exhibit 14.4. The snowflake model.

Dimensional Modelling

Having a business process

Identify all needed subjects

Who bought the products?

(customers and their structure)

Who sold the product?

(sales organization)

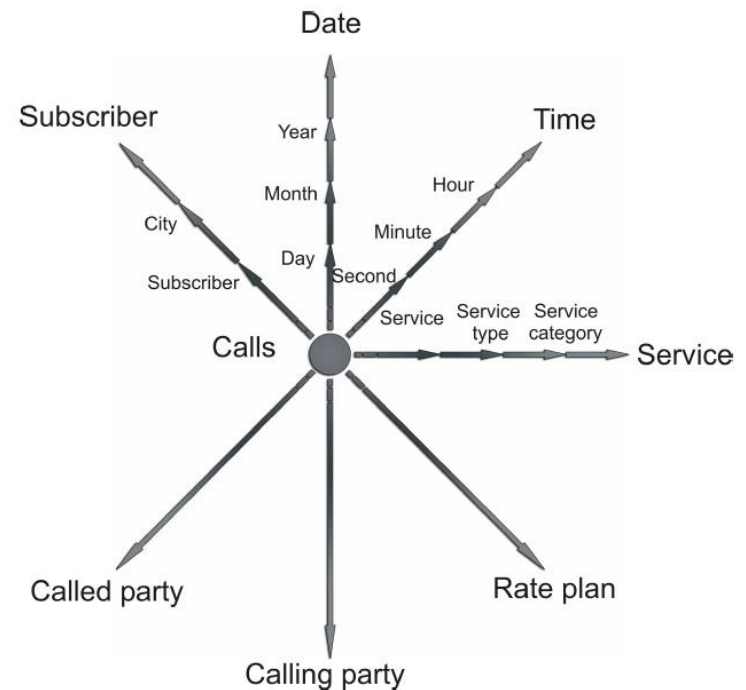
What was sold?

(product structure)

When was it sold?

(time structure)

Etc.



Dimensional Modelling

Define the measures

How the business measures success?

Measures are determined by answering the question, "What is the process measuring?"

Business users are highly interested in analyzing these performance metrics.

A single fact table row has a one-to-one relationship to a measurement event

as described by the fact table's grain.

All candidate measures in a design must be true to the grain defined.

Measures (facts) that clearly belong to a different grain must be in a separate fact table.

Typical measures are numeric additive figures

such as quantity ordered or dollar cost amount.

Dimensional Modelling

You can identify measures by looking for things that people want to “roll up” or “drill in”

Measures tend to be numeric in value, and people want to see them at various levels of detail

“What are order dollars by product category for January?”

Elements that are aggregated, summarized, or subtotaled are measures

Measures depend on context

measurements (facts) and context descriptors (dimensions)

every dimensional design sorts information requirements into these categories.

Not everything that is numeric is a measure.

Sometimes, a numeric data element is really a dimension.

Dimensional Modelling

Facts

Not everything that is numeric is a measure.

The key is to determine how it is being used.

“Show me margin dollars by order number,”

Margin dollars can be specified at various levels of detail.

Order number is being used to specify the context for margin dollars – order number is a dimension.

Other examples of numeric data elements behaving as dimensions include

size, ages, phone numbers, document numbers, and unit amounts such as unit cost or unit price.

The numbers themselves are not rolled up or down according to the question

Instead, they specify the context by which something else may be rolled up or down.

Dimensional Modelling

How to identify facts and dimensions?

is it something that can be specified at varying levels of detail?

If so, it is a fact.

is it providing context?

If so, it is a dimension.

Sorting them into categories for facts and dimensions is easy, once you know what to look for.

The clue that these elements are dimensions lies in their use.

Dimensional Modelling

Facts and dimensions can also be recognized by considering how they would be used if they were stored in a database and shown on a report:

Dimensions serve:

as “filters,” or “query predicates”

to specify groupings or “break levels” or to identify levels of subtotals.

in controlling the aggregation of measurements.

Notice, too, that dimension values may be used to sort data.

Measure serve

as elements that are aggregated, summarized, or subtotalled

These features are clues

Dimensional Modelling

What information can be gained from existing reports?

Dimensions?

Measures?

Order Report Western Region January 2009 (cont'd)					
Category	Product	SKU	Quantity Sold	Cost	Order Dollars
Packaging	Box - Large	011-4822	700	\$ 950.53	\$ 1,100.00
	Box - Medium	011-4899	1,250	\$ 1,001.84	\$ 1,380.00
	Box - Small	011-5744	1,200	\$ 1,200.72	\$ 1,330.00
	Clasp Letter	011-1729	400	\$ 352.82	\$ 356.00
	Envelope #10	021-0011	2,000	\$ 2,017.46	\$ 2,080.00
	Envelope Bubble	021-0012	1,200	\$ 866.51	\$ 1,212.00
	<i>All Packaging</i>			\$ 6,389.88	\$ 7,458.00
Pens	Gel Pen Black	017-1999	5000	\$ 116.39	\$ 120.00
	Gel Pen Blue	017-2444	2990	\$ 600.88	\$ 624.91
	Silver Pen	017-3001	50	\$ 128.46	\$ 130.00
	<i>All Pens</i>			\$ 845.73	\$ 874.91
Grand Total				\$207,229.42	\$214,896.91
Page 10 of 10					

Dimensional Modelling

Order Report Western Region January 2009 (cont'd)					
Category	Product	SKU	Quantity Sold	Cost	Order Dollars
Packaging	Box - Large	011-4822	700	\$ 950.53	\$ 1,100.00
	Box - Medium	011-4899	1,250	\$ 1,001.84	\$ 1,380.00
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Page 10 of 10					

Dimensional Modelling

Order Report Western Region January 2009 (cont'd)					
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Page 10 of 10					

Dimensional Modelling

Order Report Western Region January 2009 (cont'd)					
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	Envelope Bubble	021-0012	1,200	\$ 866.51	\$ 1,212.00
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	Gel Pen Blue	017-2444	2990	\$ 600.88	\$ 624.91
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	<i>All Pens</i>			\$ 845.73	\$ 874.91
Grand Total				\$207,229.42	\$214,896.91
Page 10 of 10					

Dimensional Modelling

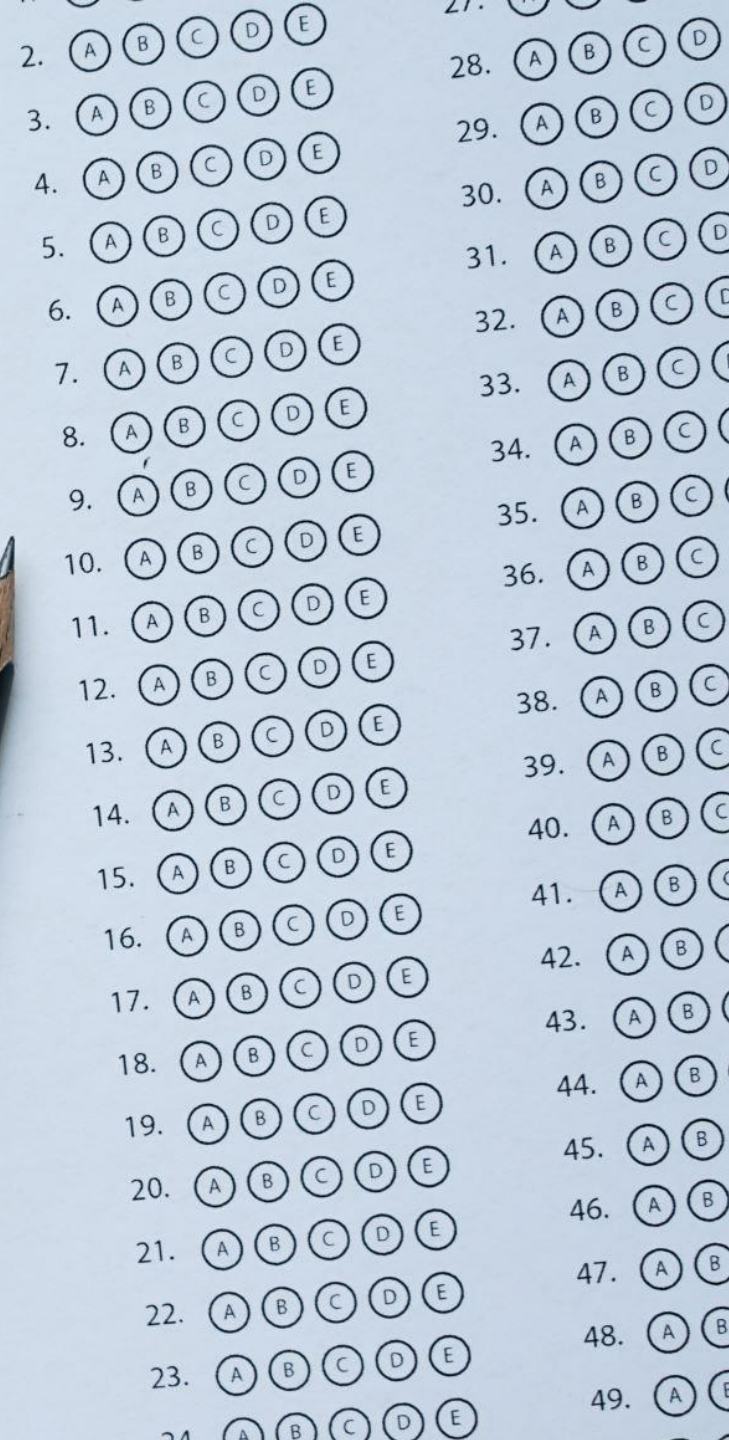
In some cases, it can be hard to tell whether a data element is a measure or a dimension.

- Sometimes, a single data element can be used in both ways.

- It is also possible to find a process where there does not appear to be a clear measurement.

- Some important business metrics are not so easily “rolled up”.

Very Basic example



Example:

The business objective is to create a data model that can store and report number of burgers and fries sold from a specific McDonalds outlet per day.

Step 0: Identify the business process

Events which snapshots are further stored within data warehouse

Event represents a sale

Sales

Example:

The business objective is to create a data model that can store and report number of burgers and fries sold from a specific McDonalds outlet per day.

Step 1: Identify the dimensions

Dimensions are the object or context.

3 different things - some "food", some specific McDonalds "store" and some specific "day"

Example:

The business objective is to create a data model that can store and report number of burgers and fries sold from a specific McDonalds outlet per day.

Step 2: Identify the measures

Measures are the quantifiable subjects and these are often numeric in nature.

The number of burgers/fries sold is a measure.

Example:

The business objective is to create a data model that can store and report number of burgers and fries sold from a specific McDonalds outlet per day.

Step 3: Identify the attributes or properties of dimensions

We need to know what are the properties or attributes of each dimension

We need to check which of these attributes are relevant

which of available attributes are required for reporting on this data.

As for the given statement above, we just need to know only one attribute of the "food" dimension - i.e. name of the food (burger or fries)

Example:

The business objective is to create a data model that can store and report number of burgers and fries sold from a specific McDonalds outlet per day.

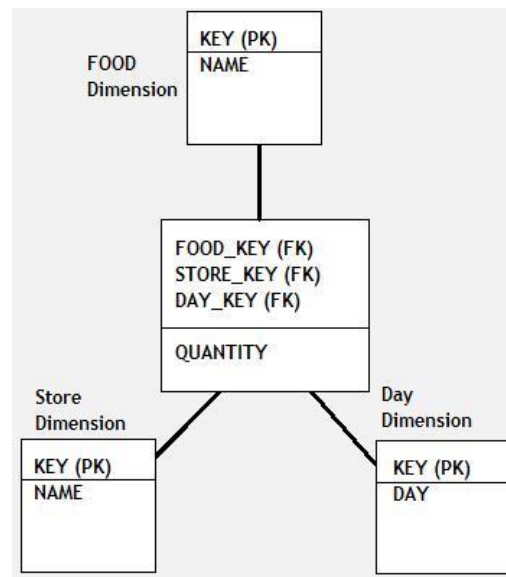
Step 4: Identify the granularity of the measures

Granularity refers to the lowest (or most granular) level of information stored in any table

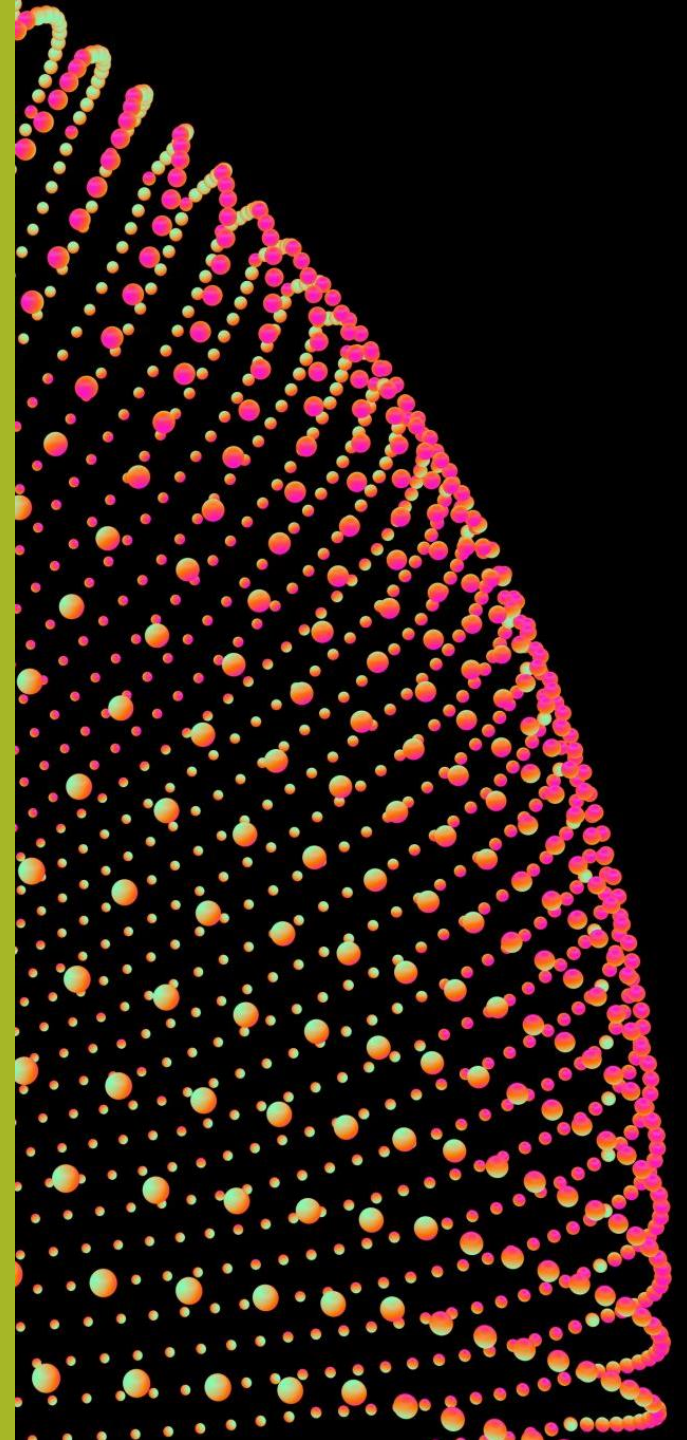
In our case, we need the information on a daily basis

Example:

The business objective is to create a data model that can store and report number of burgers and fries sold from a specific McDonalds outlet per day.



Dot modelling



Conceptual Model

Dot modelling is based on the simplified requirements for dimensional modelling

The method was developed as a kind of evolution using dimensional concepts and has been evolved to adapt to the requirements of the customer centric approach

The model is composed of three basic components:

Dot, which represents the facts

The information represented by the dot is usually numeric. It is either atomic or complex value

Dimension names.

several dimensions by which the information represented by the dot may be analyzed, the hierarchal structure of a dimension is represented by arrows

Connectors, which are placed between the facts and dimensions.

Connectors show first-level dimensions. Similarly, connectors are placed between dimensions and groupings to show the hierarchical structure

Conceptual Model

The method uses a set of worksheets.

The first worksheet is the data model worksheet itself

It contains the following:

Name of the application, or model

Diagram

List of the fact attributes (i.e., quantity and value)

Conceptual Model

Start with a dot

we start by modeling behavior

*dot is not an acronym - it comes from
the characteristic that the center of
the behavioral part of the model,
the facts, are represented by a dot.*

The information represented by the
dot is usually numeric.

*It could be an atomic value,
such as the monetary value of the sale,
or it could be complex and may
include other values
such as the unit quantity and profit on
the sale*



Dimensional Modelling

Choose business process

Goal

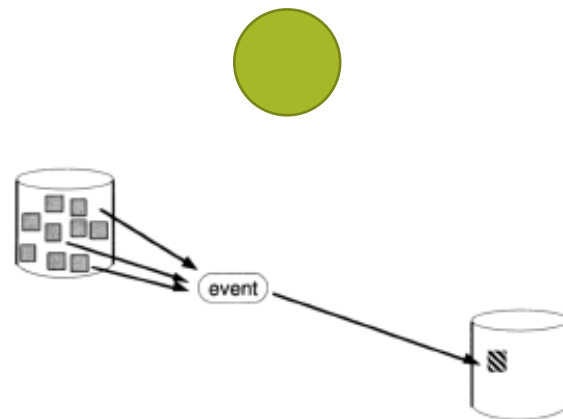
Define the purpose, and decide on the subject(s) for the data warehouse

Identify questions of interest

Snapshots

the snapshot has been taken as a result of an event

The first step in designing the data warehouse is to identify the events

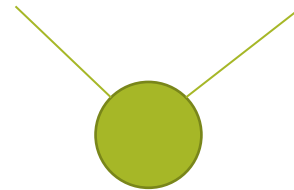


Conceptual Model

Further you add dimensions by which the information represented by the dot may be analyzed

- dot will continue to represent the same information irrespective of how many dimensions are needed to analyze and report upon it

- dimensions are arranged around the dot



Dimensional Modelling

Having a business process

Identify all needed subjects

Who bought the products?

(customers and their structure)

Who sold the product?

(sales organization)

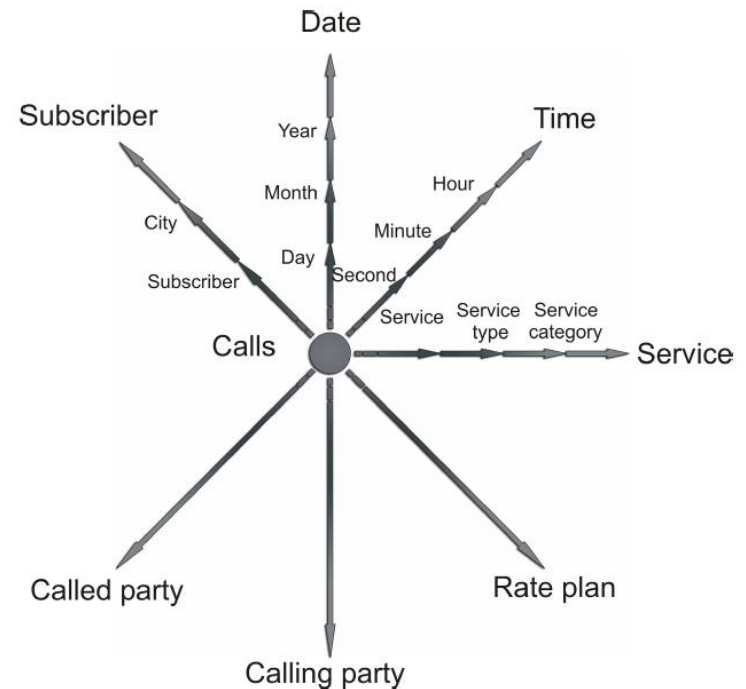
What was sold?

(product structure)

When was it sold?

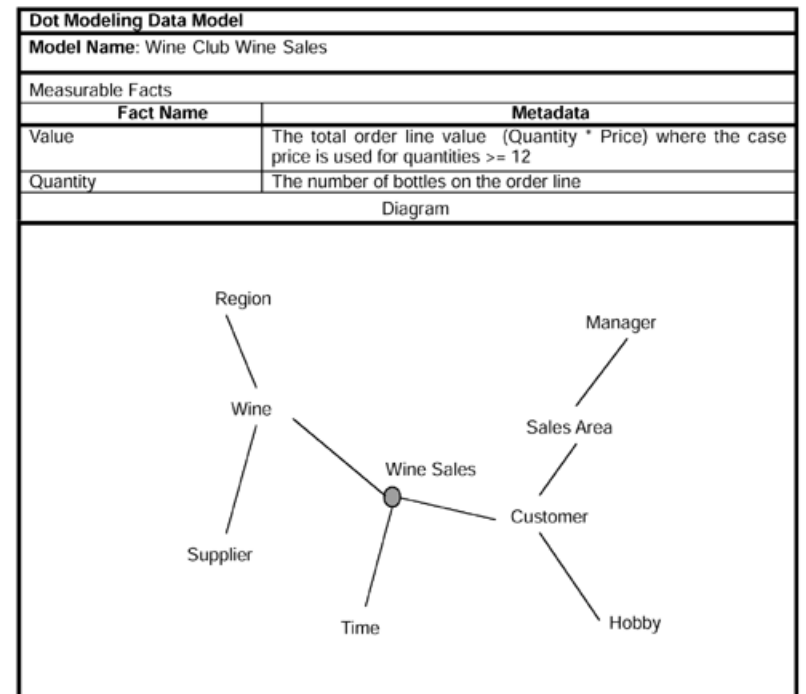
(time structure)

Etc.



Dot Model

Data model worksheet



Dot Model

A second worksheet, the entities worksheet

The purpose of the entities worksheet is to aid the designers of the system to understand the requirements in order to assist them in the logical design.

For each entity the following items of information are recorded:

Name of the dimension as it is understood by the business people.

For example, customer.

Retrospection of the entity's existence.

permanent retrospection (once set always the same)

real retrospection (new versions are separate, requires additional time mark)

false retrospection (new versions replace old ones, there is no access to previous values)

Frequency of the capture of changes to the existence of the dimension.

This will help to establish whether the dimension will be subject to errors of temporal synchronization.

Dot Model

Dot Modeling Entities and Segments		
Entity Name	Retrospection	Frequency
Customer	True	Monthly
Metadata The customer entity contains all customer details. We require to monitor discontinuous existences and to be able to identify active and non active customers. An active customer is defined to be one that has placed an order within the last twelve months. Any customer not having done so will be classified as inactive. Subsequent orders enable the customer to be re-classified as active but the periods of inactivity must be preserved.		

Dot Model

For each dimension, a set of attributes is also defined on a separate worksheet refers to the properties of other attributes.

For each attribute, the following information is recorded:

- Name of the dimension that owns it.

- Name of the attribute (name as the business (nontechnical) people would refer to it)

- Retrospection (whether or not the historical values of this attribute should be faithfully recorded)

- Frequency (frequency with which the data is recorded in the data warehouse)

 - This is an important component in the determination of the accuracy of the data warehouse.*

- Dependency (This relates to causality and identifies other attributes that this attribute is dependent upon)

- Identifying attribute (indicates whether the attribute is the identifying attribute).

- Metadata (business description of the attribute).

- Source (mapping back to the source system, where the attribute actually comes from).

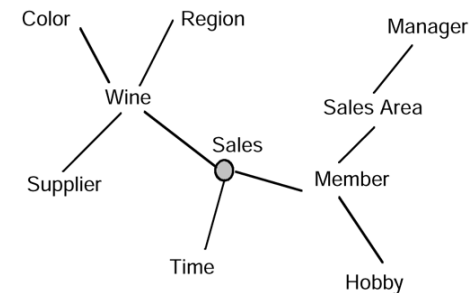
- Transformations (Any processing that must be applied to the attribute before it is eligible to be brought into the data warehouse).

 - Examples - the restructuring of dates to the same format, the substitution of default values in place of nulls or blanks.*

- Data type (data type, and precision of the attribute).

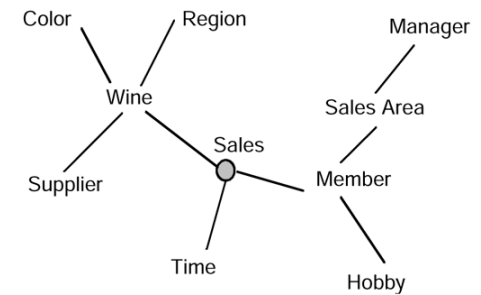
Dot Model

Dot Modeling Entities and Segments			
Entity Name		Retrospection	Frequency
Customer		True	Monthly
Metadata The customer entity contains all customer details. We require to monitor discontinuous existences and to be able to identify active and non active customers. An active customer is defined to be one that has placed an order within the last twelve months. Any customer not having done so will be classified as inactive. Subsequent orders enable the customer to be re-classified as active but the periods of inactivity must be preserved.			
Attribute Name		Customer Code	
		PK? Y	
Retrospection		Frequency	Dependency
Permanent		Monthly	None
Metadata: The identifier for Customers. This Code is supplied by the operational system.			
Source		Transformation	Data type
Customer Admin		None	Number 4
Attribute Name		Customer Name	
		PK? N	
Retrospection		Frequency	Dependency
False		Monthly	None
Metadata: The name of the customer in the form of surname preceded by initials			
Source		Transformation	Data type
Customer Admin		None	Char 25
Attribute Name		Customer Address	
		PK? N	
Retrospection		Frequency	Dependency
True		Monthly	Sales Area hierarchy
Metadata: The customer's address			
Source		Transformation	Data type
Customer Admin		None	Char 75
Attribute Name		Date Joined	
		PK? N	
Retrospection		Frequency	Dependency
Permanent		N/A	None
Metadata: This is the date the customer joined the club.			
Source		Transformation	Data type
Customer Admin		Standard date conversion	Date



Dot Model

Dot Modeling Entities and Segments			
Entity Name		Retrospection	Frequency
Manager		True	Monthly
Metadata			
The manager dimension contains details of the managers of the sales areas.			
Attribute Name	Manager Code		PK? Y
Retrospection	Frequency	Dependency	
Permanent	Monthly	None	
Metadata:			
The identifier for managers.			
Source	Transformation		Data type
Payroll	None		Char 2
Attribute Name	Manager Name		PK? N
Retrospection	Frequency	Dependency	
False	Monthly	None	
Metadata:			
Manager's Name (First Name and Last Name)			
Source	Transformation		Data type
Payroll	None		Char 25



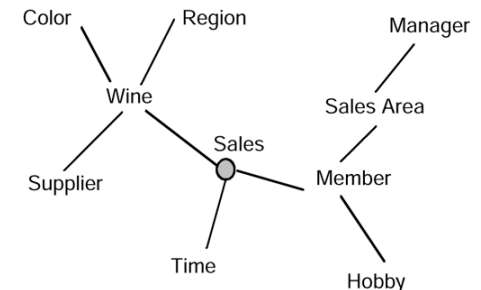
Dot Model

For each fact, fact usage worksheet is determined

Defines whether a particular dimensions is connected with a fact

Defines different aggregation methods

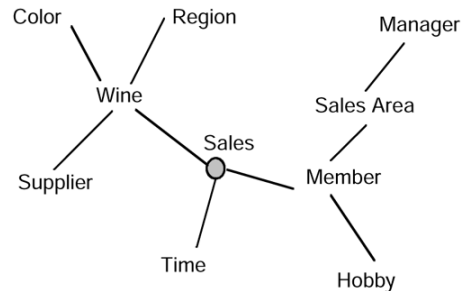
Dot Modeling Fact usage					
Model Name: Wine Sales					
Fact Name: Value		Frequency Daily			
Dimensions	Sum	Count	Ave	Min	Max
1) Manager	√	√	√	√	√
2) Sales Area	√	√	√	√	√
3) Customer	√	√	√	√	√
4) Hobby	√	√	√	√	√
5) Color	√	√	√	√	√
6) Region	√	√	√	√	√
7) Wine	√	√	√	√	√
8) Supplier	√	√	√	√	√

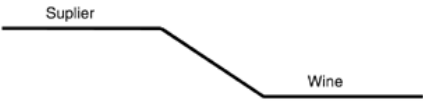


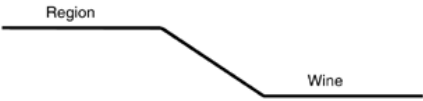
Dot Model

Finally, hierarchies and groupings worksheet is determined

Defines internal relations between attributes within dimensions



Dot Modeling Hierarchies & Groupings			
Model Name		Wine Sales	
			
Retrospection		True	Frequency
			Monthly
Metadata This hierarchy records the supply of wines to the club. Each wine may be supplied by one supplier at a time although, from time to time, the supplier for a particular wine may be changed.			

Dot Modelling Hierarchies & Groupings			
Model Name		Wine Sales	
			
Retrospection		Permanent	Frequency
			Monthly
Metadata			
This grouping records the relationship between wines and the region in which the grapes are grown.			

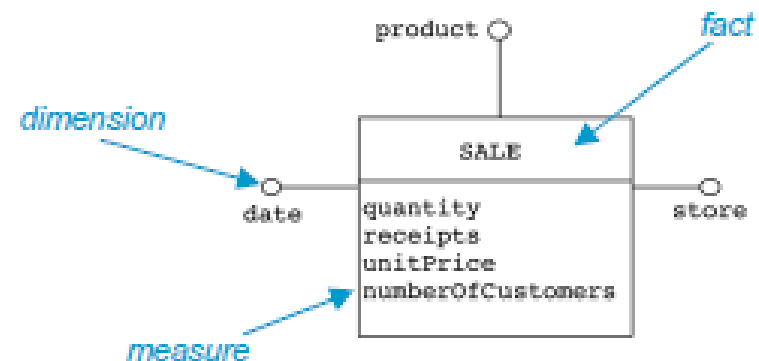
Dimensional Fact Model

Dimensional Fact Model (DFM)

a conceptual modeling approach
specifically designed for data
warehousing and business intelligence
environments

addresses the unique requirements of
analytical processing by providing a clear
and intuitive way to represent
multidimensional data structures

typically visualized as a collection of
tree-structured schemas (fact schemas),
where a central fact is connected to its
related dimensions



Fact schemas

Facts with measures

Dimensions with hierarchies

Dimensional Fact Model

All attributes and measures in the fact schema must have different names

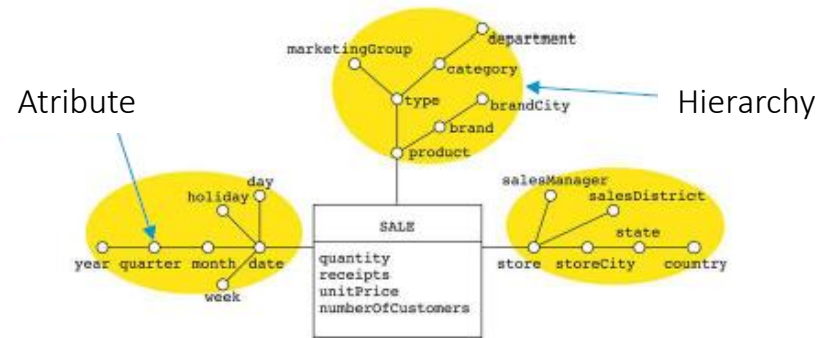
You can distinguish between similar names if you qualify them with the name of the dimension attribute that comes before them in the hierarchy

For example, storeCity and brandCity

Attribute names should not explicitly refer to the fact they belong to

Avoid “shipped product” and “shipment date”

Attributes with the same meaning in different fact schemas should have the same name



Dimensional Fact Model

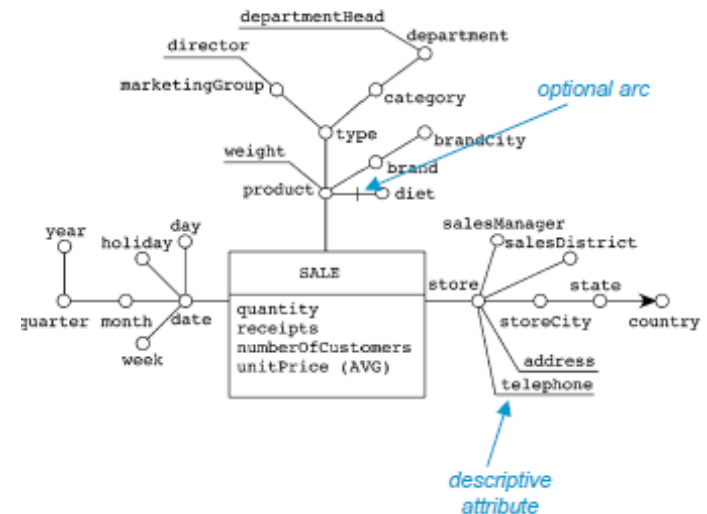
Descriptive attribute

Attributes that provide additional information about a dimension attribute but are not used for aggregation or grouping

(e.g., Telephone Number, Weight).

Enhance the informational content of dimension members; always leaves in a hierarchy.

Shown as horizontal underscore



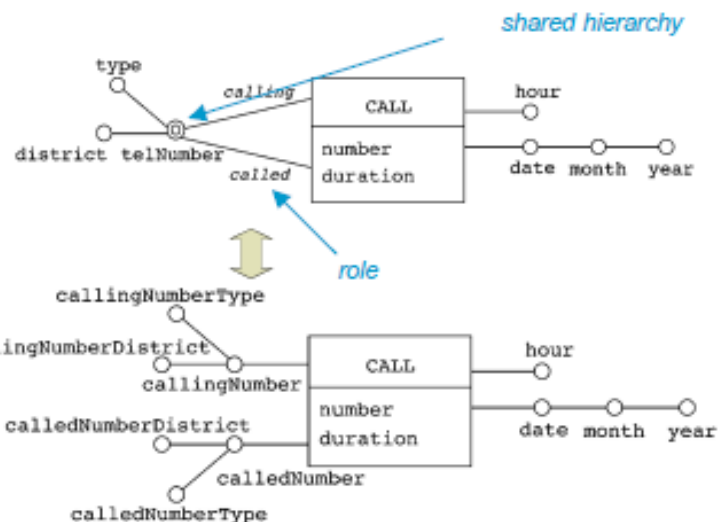
Optional attribute

Attributes that are only defined for a subset of events or entities

(e.g., Diet attribute only for food products).

Model real-world scenarios where not all attributes apply universally.

Marked with a dashed line



Shared hierarchy

Abbreviation meaning that part of the hierarchy is replicated two or more times in the fact schema.

Dimensional Fact Model

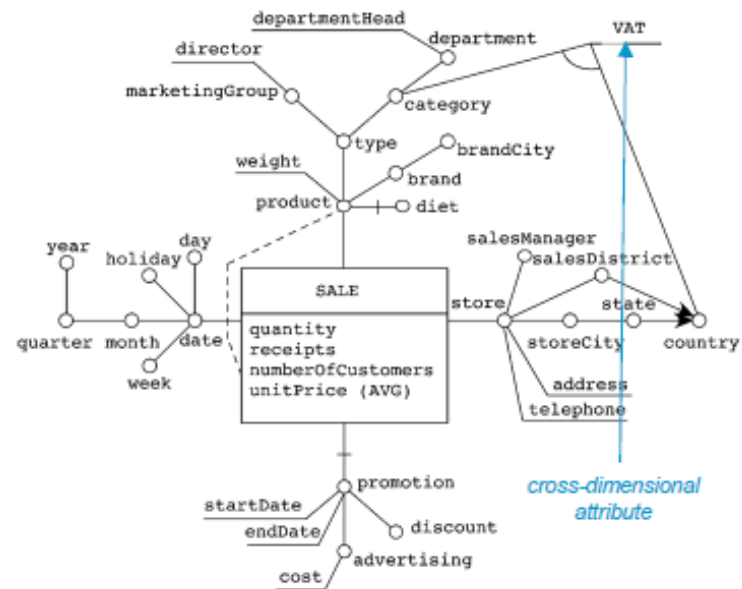
Cross-dimensional attribute

Attributes whose values depend on a combination of two or more dimension attributes, possibly from different hierarchies

(e.g., VAT rate determined by both Product Category and Country).

Capture business rules or properties that span multiple dimensions.

Illustrated with circular arcs joining the relevant dimensions



StarER

The starER model

combines the star structure that is prevalent in data warehouses, with the semantically rich constructs of the ER model

Special relationship types have been added to handle hierarchies

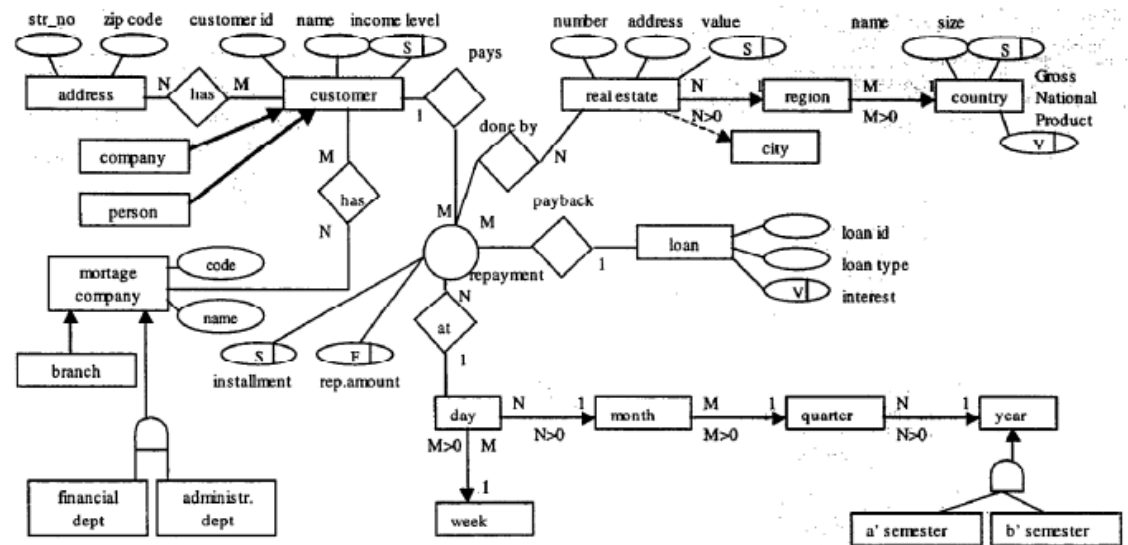


Figure 3.9: An excerpt from a mortgage company data warehouse, using the starER model. The financial and the administration department are outside the customer dimension as this is dependent on design decisions and semantic issues.

ME/R

Specialization of E/R model
called ME/R

In order to express a
multidimensional data structure, it
defines

two specialized sets of relations
and a specialized set of entities.

The resulting ME/R model allows
for an appropriate conceptual
representation of the
multidimensional view of data

inherent in OLAP

*separation of defining/descriptive
and quantitative data allows for a
complex structure of dimensions*

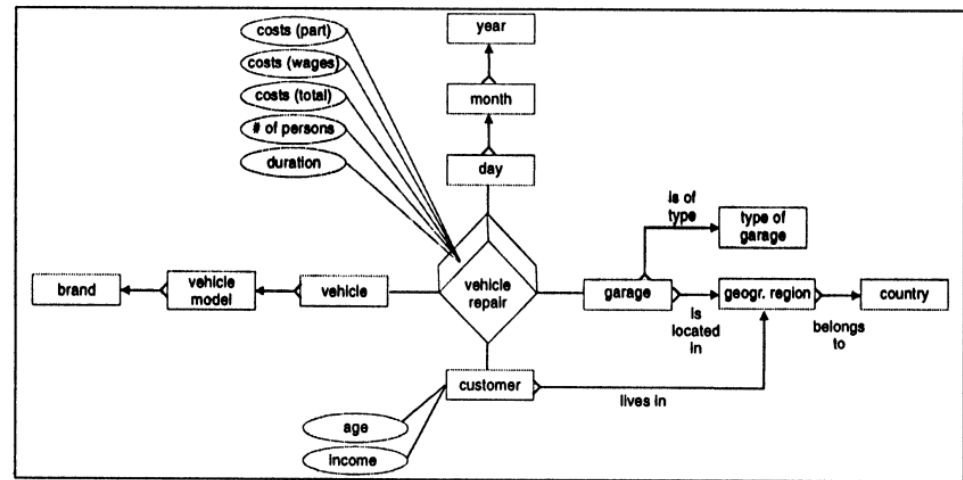
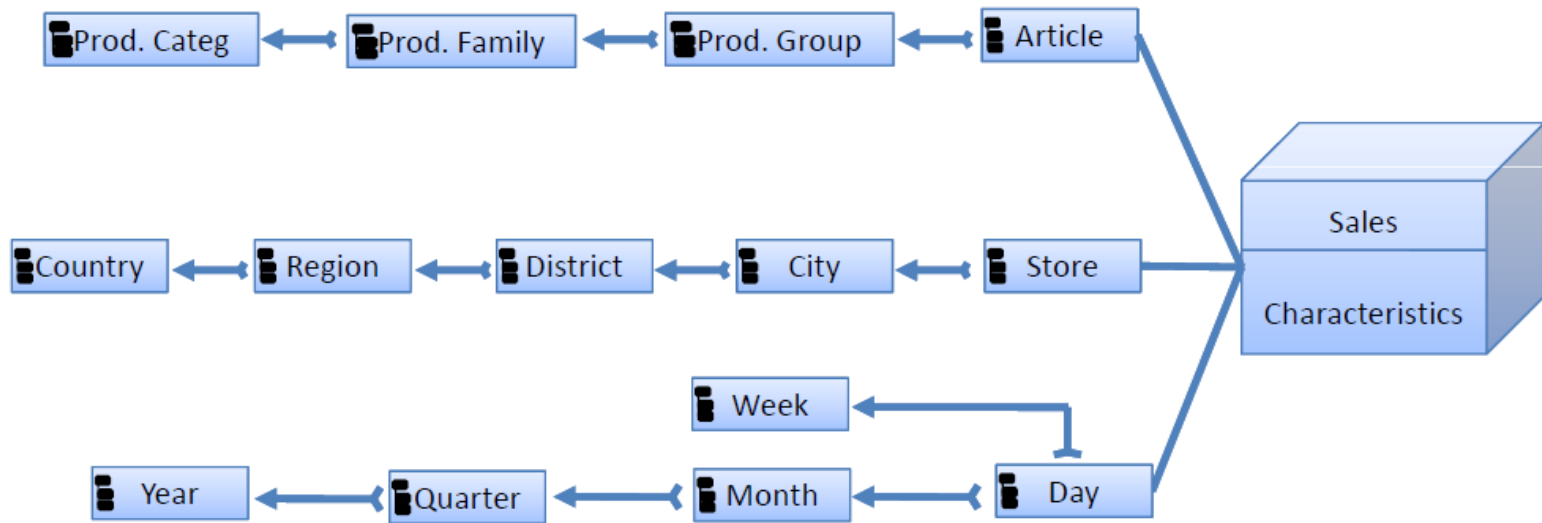
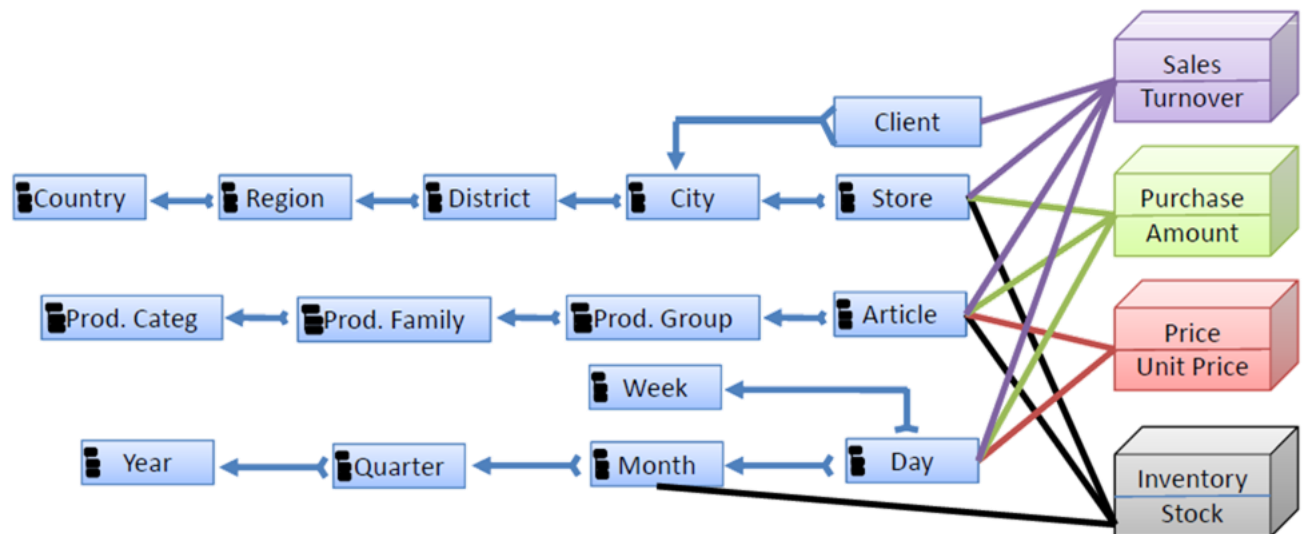


Fig. 7. The ME/R diagram for the analysis of vehicle repairs



ME/R



Logical Data Map



Logical Map

Before you begin building your extract systems,
you need a logical data map

documents the relationship between original source fields and final destination fields in the tables you deliver to the front room
ties the very beginning of the ETL system to the very end

The physical implementation can be a catastrophe
if it is not carefully architected before it is implemented
just as with any other form of construction
you must have a blueprint before you hit the first nail

Logical Map

Example:

Information about sales

where is the sale in DB

It is a common problem:

that there is no table SALES in the operational databases

Though, some other tables can exist like ORDER with an attribute ORDER STATUS.

that there is no table Time in the operational databases

Though, some tables may contain DATE type attributes

Logical Map

Logical Data Map

actual design of the logical data mapping document

provide the ETL developer with a clear-cut blueprint of exactly what is expected from the ETL process

connects the original source data to the final data

a map for the extraction process

contains

the data definition for the data warehouse source systems

the target data warehouse data model

the exact manipulation of the data required to transform it from its original format to that of its final destination.

Logical Map

Components of the Logical Data Map

The logical data map is usually presented in a table or spreadsheet format:

Target					Source				Transformation
Table Name	Column Name	Data Type	Table Type	SCD Type	Database Name	Table Name	Column Name	Data Type	
EMPLOYEE_DIM	EMPLOYEE_KEY	NUMBER	Dimension	1				NUMBER	Surrogate key.
EMPLOYEE_DIM	EMPLOYEE_ID	NUMBER	Dimension	1	HR_SYS	EMPLOYEES	EMPLOYEE_ID	NUMBER	Natural Key for employee in HR system
EMPLOYEE_DIM	BIRTH_COUNTRY_NAME	VARCHAR2(75)	Dimension	1	HR_SYS	COUNTRIES	NAME	VARCHAR2(75)	select c.name from employees e, states s, countries c where e.state_id = s.state_id and s.country_id = c.country
EMPLOYEE_DIM	BIRTH_STATE	VARCHAR2(75)	Dimension	1	HR_SYS	STATES	DESCRIPTION	VARCHAR2(255)	select s.description from employees e, states s where e.state_id = s.state_id
EMPLOYEE_DIM	DISPLAY_NAME	VARCHAR2(75)	Dimension	1	HR_SYS	EMPLOYEES	FIRST_NAME	VARCHAR2(75)	select initcap(salutation) ' ' initcap(first_name) ' ' initcap(last_name) from employee
EMPLOYEE_DIM	BIRTH_DATE	DATE	Dimension	1	HR_SYS	EMPLOYEES	DOB	DATE	trunc(DOB)
EMPLOYEE_DIM	SALUTATION	VARCHAR2(12)	Dimension	1	HR_SYS	EMPLOYEES	SALUTATION	VARCHAR2(12)	initcap(salutation)
EMPLOYEE_DIM	FIRST_NAME	VARCHAR2(30)	Dimension	1	HR_SYS	EMPLOYEES	FIRST_NAME	VARCHAR2(30)	initcap(first_name)
EMPLOYEE_DIM	LAST_NAME	VARCHAR2(30)	Dimension	1	HR_SYS	EMPLOYEES	LAST_NAME	VARCHAR2(30)	initcap(last_name)
EMPLOYEE_DIM	MARITAL_STATUS	VARCHAR2(12)	Dimension	2	HR_SYS	MARITAL_STATUS	DESCRIPTION	VARCHAR2(12)	select nvl(m.name,'Unknown') from employee e marital_status m where e.marital_status_id = m.marital_status_id
EMPLOYEE_DIM	DIVERSITY_CATEGORY	VARCHAR2(30)	Dimension	1	HR_SYS	EMPLOYEES	EEO_CLASS	VARCHAR2(30)	decode(eeo_class,null, 'Not Stated', decode(eeo_class,'N', 'Not Stated',eeo_class))
EMPLOYEE_DIM	GENDER	VARCHAR2(12)	Dimension	1	HR_SYS	EMPLOYEES	SEX	VARCHAR2(12)	nvl(sex, 'Unknown')
EMPLOYEE_DIM	EMPLOYEE_STATUS	VARCHAR2(24)	Dimension	1	HR_SYS	EMPLOYEES	STATUS	VARCHAR2(24)	select es.name from employee e employee_status es where e.employee_status_id = m.employee_status_id
EMPLOYEE_DIM	POSITION_CODE	VARCHAR2(12)	Dimension	2	HR_SYS	POSITIONS	POSITION_CODE	VARCHAR2(12)	select p.code from employees e, positions p where p.position_id = e.position_id
EMPLOYEE_DIM	POSITION_CATEGORY	VARCHAR2(30)	Dimension	2	HR_SYS	POSITIONS	POSITION_CATEGORY	VARCHAR2(30)	select p.category from employees e, positions p where p.position_id = e.position_id
EMPLOYEE_DIM	HIRE_DATE	DATE	Dimension	1	HR_SYS	EMPLOYEES	DATE_HIRED	DATE	trunc(date_hired)
									select d.code from employee p, employee_department pd, departments d where p.employee_id= pd.employee_id and

Logical Map

Components of the Logical Data Map

The logical data map is usually presented in a table or spreadsheet

Target				
Table Name	Column Name	Data Type	Table Type	SCD Type
EMPLOYEE_DIM	EMPLOYEE_KEY	NUMBER	Dimension	1
EMPLOYEE_DIM	EMPLOYEE_ID	NUMBER	Dimension	1
EMPLOYEE_DIM	BIRTH_COUNTRY_NAME	VARCHAR2(75)	Dimension	1

Source			
Database Name	Table Name	Column Name	Data Type
			NUMBER
HR_SYS	EMPLOYEES	EMPLOYEE_ID	NUMBER
HR_SYS	COUNTRIES	NAME	VARCHAR2(75)

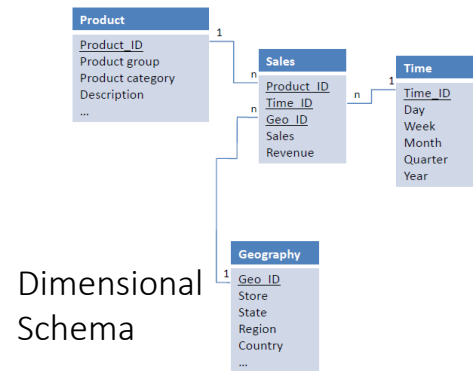
Transformation
Surrogate key.
Natural Key for employee in HR system
select cname from employees e, states s, countries c where e.state_id = s.state_id and s.country_id = c.country

Logical map

How to prepare Logical Map:

We start with .. ?

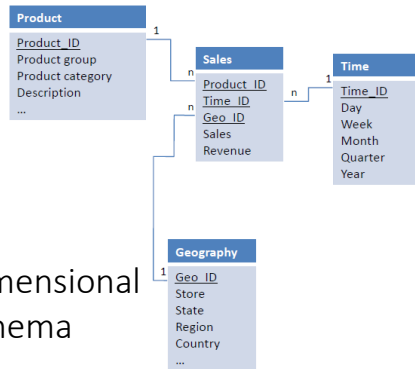
We finish with .. ?



Data
Sources



Logical Map



Target				
Table Name	Column Name	Data Type	Table Type	SCD Type
EMPLOYEE_DIM	EMPLOYEE_KEY	NUMBER	Dimension	1
EMPLOYEE_DIM	EMPLOYEE_ID	NUMBER	Dimension	1
EMPLOYEE_DIM	BIRTH_COUNTRY_NAME	VARCHAR2(75)	Dimension	1

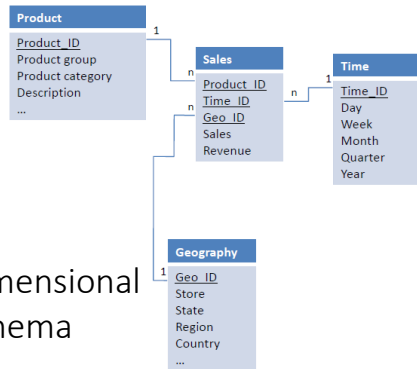


Source			
Database Name	Table Name	Column Name	Data Type
			NUMBER
HR_SYS	EMPLOYEES	EMPLOYEE_ID	NUMBER
HR_SYS	COUNTRIES	NAME	VARCHAR2(75)

Dimensional
Schema

Data
Sources

Logical Map



Dimensional Schema

Target				
Table Name	Column Name	Data Type	Table Type	SCD Type
EMPLOYEE_DIM	EMPLOYEE_KEY	NUMBER	Dimension	1
EMPLOYEE_DIM	EMPLOYEE_ID	NUMBER	Dimension	1
EMPLOYEE_DIM	BIRTH_COUNTRY_NAME	VARCHAR2(75)	Dimension	1

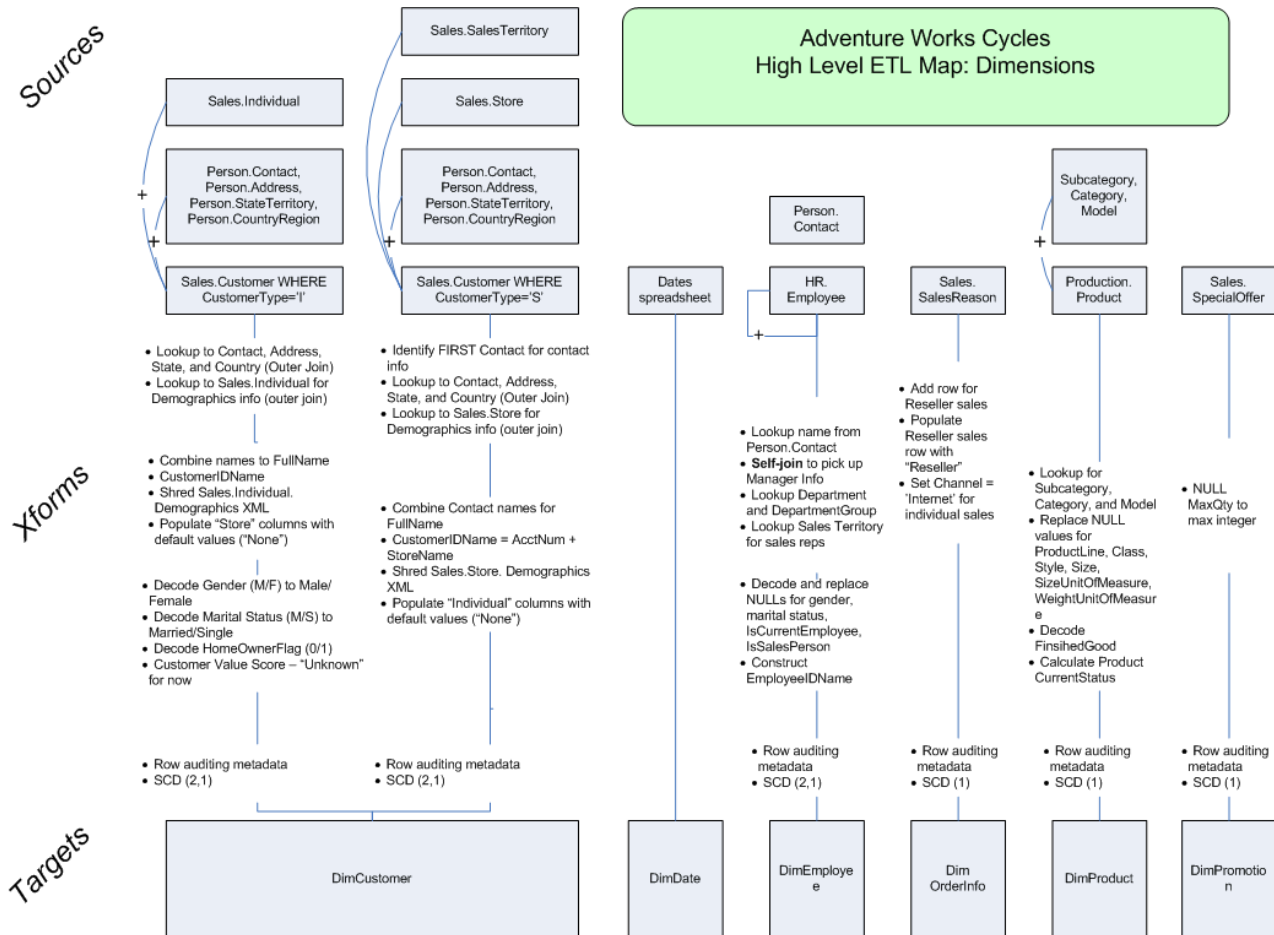
Transformation
Surrogate key.
Natural Key for employee in HR system
select c.name from employees e, states s, countries c where e.state_id = s.state_id and s.country_id = c.country
select e.description from employees e, states s

Source			
Database Name	Table Name	Column Name	Data Type
			NUMBER
HR_SYS	EMPLOYEES	EMPLOYEE_ID	NUMBER
HR_SYS	COUNTRIES	NAME	VARCHAR2(75)



Data Sources

High level ETL map

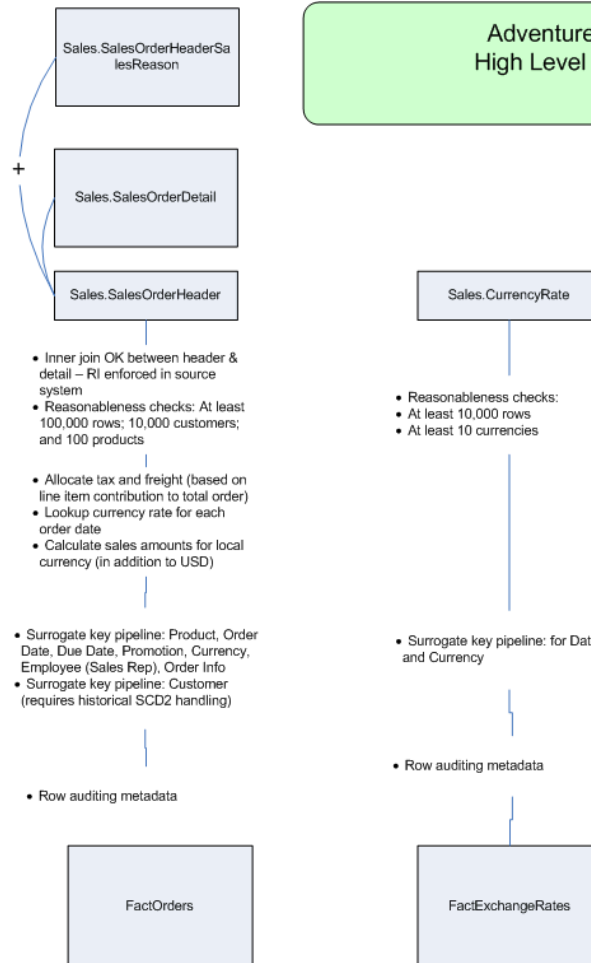


High level ETL map

Sources

Xforms

Targets



ETL

Design

Always Logical Before Physical

Have a plan

The ETL process must be figured out logically and documented

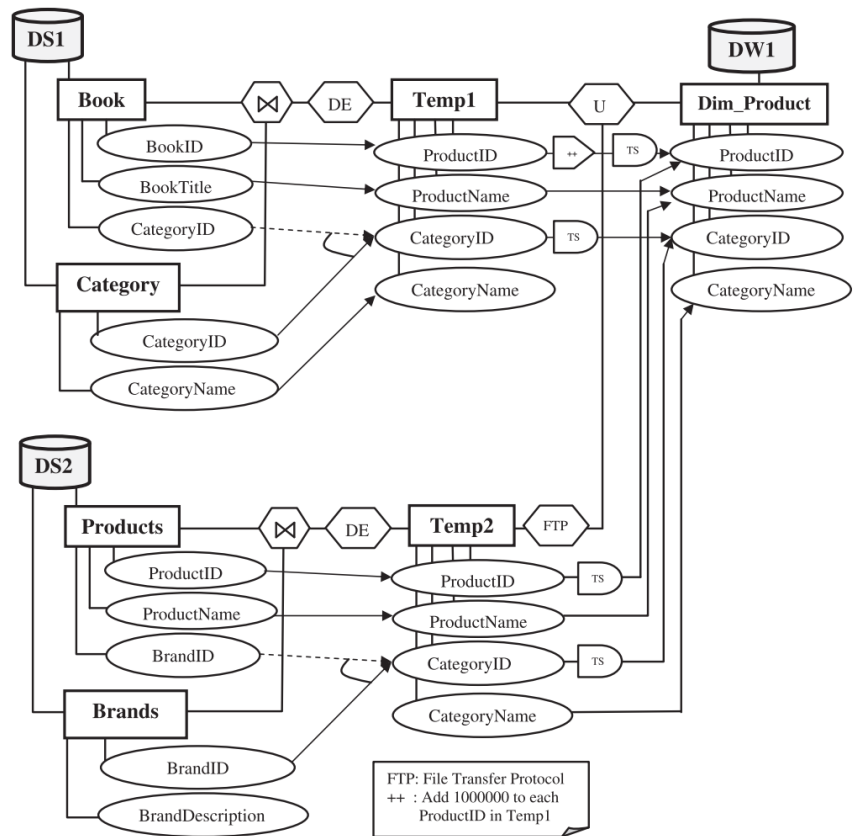


Figure 16 EMD scenario for building products dimension.

ETL

Design (continued):

Identify data source candidates

Starting with the highest-level business objectives

identify the likely candidate data sources you believe will support the decisions needed by the business community

identify specific data elements you believe are central to the end user data

These data elements are then the inputs to the data profiling step

Analyze source systems with a data-profiling tool

Data in the source systems must be scrutinized for data quality and completeness

ETL

Designing Logical Before Physical (continued):

Receive walk-through of data lineage and business rules

The data-profiling step should have created two subcategories of ETL-specific business rules:

- Required alterations to the data during the data-cleaning steps

- Coercions to dimensional attributes and measured numerical facts
to achieve standard conformance across separate data sources

Receive walk-through of data warehouse data model

must have a thorough understanding of how dimensions, facts, and other special tables in the dimensional model work together

Validate calculations and formulas

*Verify with end users any calculations specified in the data lineage
measure twice, cut once*

ETL

Dimensional Data Structures

Dimensional data structures are the target of the ETL processes, and they sit at the boundary between the back room and the front room

In many cases, the dimensional tables will be the final physical-staging step before transferring the tables to the end user environments

Fact tables

Dimension tables

Surrogate key mapping tables

ETL

Logical Design First

1. Have a (preliminary) plan
2. Identify data source candidates
3. Analyze source systems with a data-profiling tool
Possible STOP here
4. Receive walk-through of data lineage and business rules
Required alterations to the data during the data-cleaning steps
Coercions to dimensional attributes and measured numerical facts to achieve standard conformance across separate data sources
5. Receive walk-through of data warehouse data model
6. Validate calculations and formulas (*measure twice, cut once*)

Staging

Stage

typically transformed data is not directly loaded into the target data warehouse

this data should first enter a staging database

making it easier to roll back if something goes wrong

at this stage it is easier to

generate audit reports for regulatory compliance

diagnose and repair data problems

Staging

The back-room area of the data warehouse has frequently been called the staging area.

- Staging in this context means writing to disk

- The staging area stores data on its way to the final presentation area of the data warehouse.

It is recommended to stage data at the four major checkpoints of the ETL data flow:

- Extract → Clean → Conform → Deliver

To stage your data or not depends on two conflicting objectives:

- Getting the data from the originating source to the ultimate target as fast as possible

- Having the ability to recover from failure without restarting from the beginning of the process

Staging

When the staging area is initially set up

the ETL architect must supply an overall data storage measure of the staging area

estimate the space allocations

and parameter settings for the staging database, file systems, and directory structures

staging area volumetric worksheet

focusing on the final delivery tables at the end of the ETL data flow

Heterogeneous Staging Area

Heterogeneous source systems may call for a heterogeneous data staging area

Staging

Staging area volumetric worksheet

Table Name	Update Strategy	Load Frequency	ETL Job(s)	Initial Rowcount	Avg Row Length	Grows with	Expected Monthly Rows	Expected Monthly Bytes	Initial Table Size Bytes	Table Size 6 mo. (MB)
S_ACCOUNT	Truncate / Reload	Daily	SAccount	39,933	27	New accounts	9,983	269,548	1,078,191	2.57
S_ASSETS	Insert / Delete	Daily	SAssets	771,500	78	New assets	192,875	15,044,250	60,177,000	143.47
S_BUDGET	Truncate / Reload	Monthly	SBudget	39,932	104	Refreshed monthly	9,983	1,038,232	4,152,928	9.90
S_COMPONENT	Truncate / Reload	On demand	SComponent	21	31	Components added to Inventory	5	163	651	0.00
S_CUSTOMER	Truncate / Reload	Daily	SCustomer	38,103	142	New customers added daily	9,526	1,352,657	5,410,626	12.90
S_CUSTOMER_HISTORY	Truncate / Reload	Daily	SCustomerHistory	2,307,707	162	Refresh with each bulk load	576,927	93,462,134	373,848,534	891.32
S_CUSTOMER_TYPE	Truncate / Reload	On demand	SCustomerType	5	21	New customer types	1	26	105	0.00
S_DEFECT	Truncate / Reload	On demand	SDefect	84	27	New defect names	21	567	2,268	0.01
S_DEFECTS	Insert Only	Daily	SDefects	8,181,132	132	Transaction defects	2,045,283	269,977,356	1,079,909,424	2,574.70
S_DEPARTMENT	Truncate / Reload	On demand	SDepartment	45	36	Departments established	11	405	1,620	0.00
S_FACILITY	Truncate / Reload	Daily	SFacility	45,260	32	New or changed facilities worldwide	11,315	362,080	1,448,320	3.45
S_HISTORY_FAIL_REASON	Insert / Delete	On demand	SHistoryFailReason	6	27	New failure codes	2	41	162	0.00
S_OFFICE	Truncate / Reload	Daily	SOffice	14	56	New offices opened	4	196	784	0.00
S_PACKAGE_MATL	Truncate / Reload	Daily	SPackageMatl	54	18	New packaged material categories	14	243	972	0.00
S_PRODUCT	Truncate / Reload	Daily	SProduct	174,641	73	New products	43,660	3,187,198	12,748,793	30.40
S_PROVIDER	Truncate / Reload	On demand	SProvider	63	45	Service providers	16	709	2,835	0.01
S_REGION	Truncate / Reload	Daily	SRegion	333	37	New or changed global regions	83	3,080	12,321	0.03
S_RESPONSES	Insert Only	Daily	SResponses	5,199,095	105	Response transaction	1,299,774	136,476,244	545,904,975	1,301.54
S_SURVEY	Truncate / Reload	Daily	SSurvey	45,891	83	Survey conducted	11,473	952,238	3,808,953	9.08

Staging

Integrity Checking

It is a good practice to have integrity checks in the ETL process rather than in the staging database

The ETL process must know how to handle data anomalies in a more automatic way – it cannot simply reject all data

business rules for different data-quality scenarios

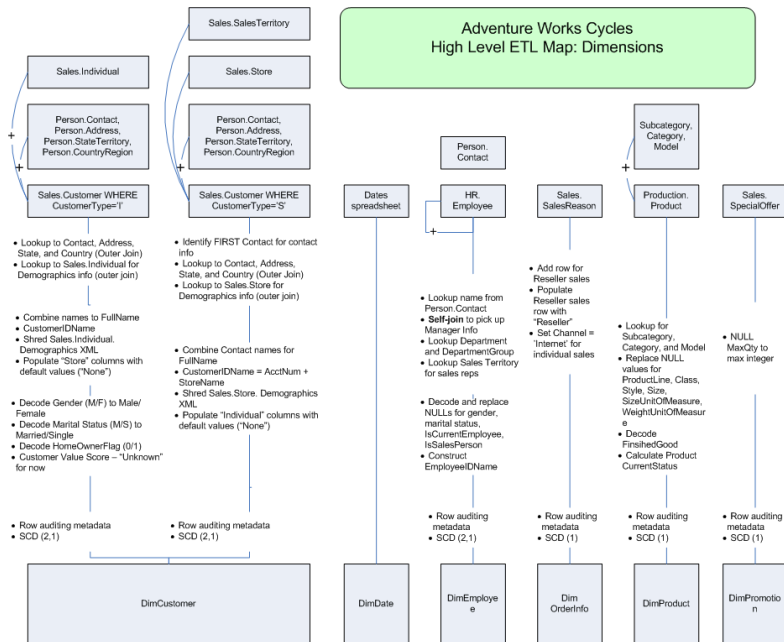
If the data is unacceptable, it has to be rejected completely and put it into a reject file for investigation.

Semi-automatic

ETL

ETL development starts out with
Preparing the logical map and high-level plan
independent of any specific technology or approach

Target DW					Data Sources				Transformation
Table Name	Column Name	Data Type	Table Type	SCD	Database Name	Table Name	Column Name	Data Type	
tf101masukpelajar	ff101kat_pelajar	Number	Fact	n/a	ASIS	t210student	f210kat_pelajar	Char(4)	Ff101kat_pelajar=f210kat_pelajar
tf101masukpelajar	ff101kdprogram	Number	Fact	n/a	ASIS	t210student	f210kdprogram	Char(4)	=
tf101masukpelajar	ff101jantina	Number	Fact	n/a	ASIS	t210student	f210jantina	Char(1)	=
tf101masukpelajar	ff101bangsa	Number	Fact	n/a	ASIS	t210student	f210bangsa	Char(1)	=
ff101masukpelajar	Ff101student_total	Number	Fact	n/a	ASIS	T210student		Char(4)	Sum(Record())
td008jantina	fd008kdjantina	Char(1)	Dimension	2	ASIS	t012jantina	f012kdjantina	Char(1)	Conversion("1", "2")
td008jantina	fd008keterangan	Char(15)	Dimension	2	ASIS	t012jantina	f012keterangan	Char(15)	
td006bangsa	fd006kdbangsa	Char(1)	Dimension	2	ASIS	t013bangsa	f013kdbangsa	Char(1)	Filter(only Malaysia)



ETL

All the dimension tables must be processed before the key lookup steps for the fact table.

The dimension tables are usually independent from each other, but sometimes they also have processing dependencies.

It's important to clarify these dependencies, as they become fixed points around which the job control flows.

ETL specification should describe

- historic load strategies for each target table

- incremental load strategies for each target table

incremental processing, by contrast, must be fully automated

Occasionally, the same ETL code can perform both the initial historic load and ongoing incremental loads

- more often you build separate ETL processes

ETL

Historic load

Start with dimensions

Start with permanent dimensions and user defined ones

Then go for type-1

Then go for other dims

Remember about

- Doing transformations – like handling nulls, decoding production codes

- Conform data

- Handle many-to-many relations

- Apply surrogate keys

- Load efficiency

Next go to fact tables

Remember about

- Handling null values

- Improving fact table – calculated measures

- Pipelining the surrogate key lookup

- Load efficiency

ETL

The steps to loading into a partitioned table include:

1. Disable foreign key (referential integrity) constraints between the fact table and each dimension table before loading data.
2. Drop or disable indexes on the fact table.
3. Load the data using fast-loading techniques.
4. Create or enable fact table indexes.
5. If necessary, perform steps to stitch together the table's partitions.
6. Confirm each dimension table has a unique index on the surrogate key column.
7. Enable foreign key constraints between the fact table and dimension tables.

ETL

Incremental load

Highly automated

Scheduling, Exception and error handling (automated for predicted, graceful for unpredicted), logging and audit

Begin with the dimension tables

Identify new and changed dimension rows

Apply proper mechanisms – depending on the SCD type

Remember about doing transformations, conforming data, handling many-to-many relations, apply surrogate keys

Next go for fact tables

Identify new and changed fact rows

Apply proper surrogate key pipelines – problem with late arriving facts

Problem with aggregates

Problem with real-time delivery



Note on data

Note on architecture

SOR – System of record

data is captured and updated in operational and transactional applications

designated as the SOR so that people and processes know what the authorized sources are for any particular data subject

SOI – System of integration

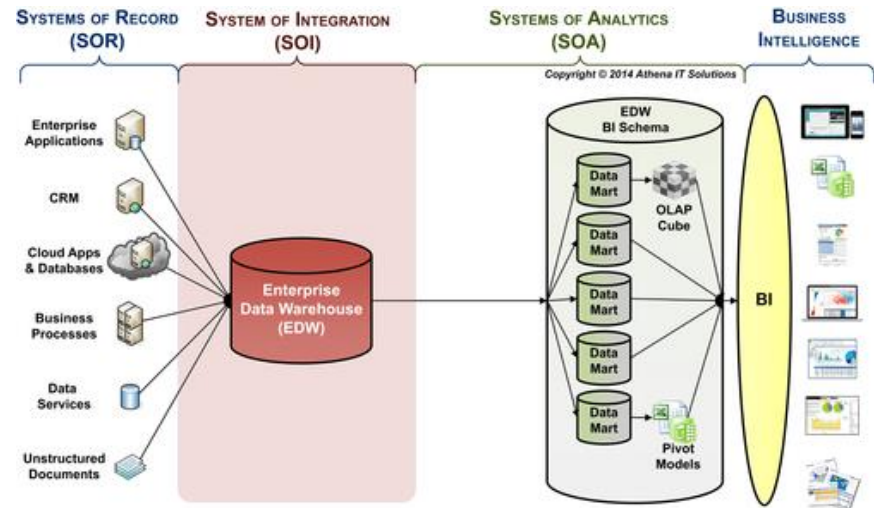
gathers, integrates, and transforms data from SORs into consistent, conformed, comprehensive, clean, and current information

implies a particular level of integrity and legitimacy of the integrated data

SOA – System of analytics

provides business information that has been integrated and transformed to BI applications for business analysis

implies a particular level of integrity and legitimacy of the information being used in BI



SORs

Business requirements need to be detailed enough to identify the data sources or systems of record (SOR)

such as applications, databases, or files needed to build the BI solution

Defining data requirements means drilling into the data sources to identify detailed data content

such as columns within tables or fields within files,

and the transformations

that are needed for consistency, conformance, or to calculate the business metrics.

In the early days of data warehousing, IT generally extracted all the data that was available in SORs in case it would be needed in the future.

Nowadays, most enterprises have too much data in their SORs to just dump it in the DW, so the BI team needs to be more judicious in selecting what is really needed for analytics

The best practice is to extract the data needed by current requirements but also to include related data

Data quality

The other area too often overlooked is examining the source systems' data quality in relation to what is being requested for analytics.

If there are gaps caused by data quality issues or the data content does not even exist, then the impact on requirements needs to be addressed.

These gaps may require:

- Identifying new sources*

- Defining data cleansing processes*

- Defining MDM (master data management) processes*

- Creating new data capture applications (outside the scope of the BI project)*

- Dropping the data requirement until it can be fulfilled*

Do not assume anything about the state of the data.

Data quality

The areas where data quality and inconsistency problems lurk include:

Data quality within SOR may be “masked” by corrections made within existing reports or spreadsheets created from this data

people using these reports or spreadsheets might not even be aware of these “adjustments”

Data does not age well

Although data quality may be fine now, there is always the chance that you will have problems or inconsistencies with the historical data.

Over time, fields are sometimes used for different content or have different business rules applied to them.

Data quality may be fine within each SOR application, but may be very inconsistent across applications

Many companies have problems with inconsistent master data in product, customer, and other dimensions that will not be apparent until the data is loaded into the enterprise DW.

Data inconsistency needs to be identified as part of the data requirements process.

Data quality

The BI team needs to perform data profiling or source systems analysis to determine the current state of data quality and its consistency within and across source systems.

Based on the data profiling results, perform a gap analysis to determine the current state of data quality within and across source systems regarding the data requirements.

With the gap analysis, determine what corrective action, if any, can be done to plug the gaps.

Review the gap analysis and potential corrective actions with the business, asking them to determine if and how the gaps affect their requirements.

The business options are to continue with the data quality as is, change their data requirements, or take some corrective action.

If they are interested in taking corrective action, then they need to assign a value to that action to determine if it is a must-have.

When the BI team revises the project plan and budget after defining the requirements, you want to identify the cost of these corrective actions and get final business sign-off.

SSoT

A critical thing that all companies need to do in order to get value out of their data is to ensure that all stakeholders are using the same data to make decisions

they need to make sure they are working from a single source of truth.

Need for managing master data

arises when there are multiple, independent sources for an entity.

Single source of truth (SSOT)

the practice of structuring information models and associated data schema such that every data element is mastered in only one place

Master Data Management

is the set of processes used to create and maintain a consistent view, also referred to as a master list, of key enterprise reference data.

This data includes such entities as customers, prospects, suppliers, employees, products, services, assets, and accounts.

It also includes the groupings and hierarchies associated with these entities.

Dimensional Modelling

What is a single source of truth?

Each system focuses on a different set of priorities and follows a different set of rules.

each system provides a partial window into company's operations

An SSOT creates a centralized copy of all the relevant data

SSOT is the practice of structuring information models and associated data schema such that every data element is mastered (or edited) in only one place

Everyone using the same standardized fields

Creating a Single Source of Truth requires data engineering effort.

For example, what is the "*close date*" for a particular purchase?

Is it the date the signed contract was submitted to DocuSign?

Is it the closed-won date in Salesforce?

Is it the date you bill the client through Stripe?

The date the payment shows up in your company's bank account?

Or when—according to your company's accounting rules—the payment shows up on your company's books?

Dimensional Modelling

What is a single source of truth (SSOT)?

For a business to obtain a SSOT

all data components from the various systems across the enterprise will aggregate their datasets to the primary single source of truth location.

To do this

*an integration strategy is needed
an interface that will host and
surface the organization's data*

Additionally, all departments and teams must provide access to the systems and data they use.

For example, what is the "close date" for a particular purchase?

Is it the date the signed contract was submitted to DocuSign?

Is it the closed-won date in Salesforce?

Is it the date you bill the client through Stripe?

The date the payment shows up in your company's bank account?

Or when—according to your company's accounting rules—the payment shows up on your company's books?

Dimensional Modelling

“Single Version of the Truth”
(SVOT)

“a commonly agreed and accepted set of truths that operate as a foundation, on which we will each derive our own, interpreted contextually centric views”

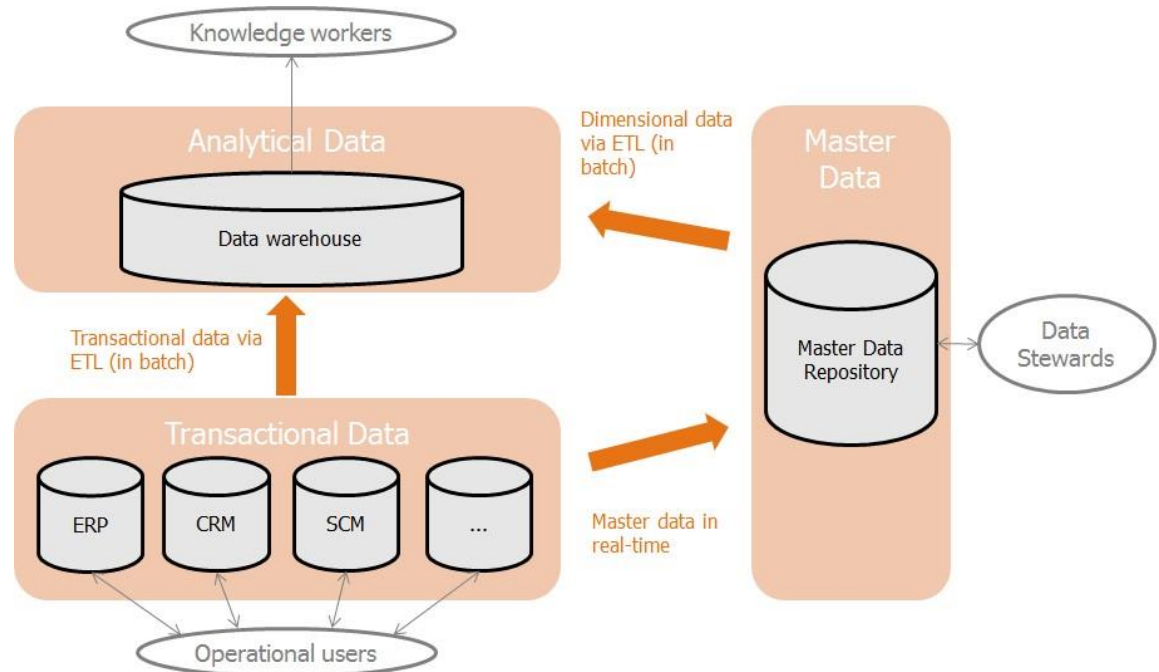
You need to have a single version of truth.

But you can have multiple versions of truth.



MDM

Many enterprises use MDM
as part of their information
architecture
to create consistent and
conformed reference data



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